List Prices in Segmented Housing Markets*

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Abstract

We study the role of list prices in frictional and segmented housing markets. Using data from a large online housing platform in Beijing, where buyers actively filter listings by preferred ranges for price and housing characteristics, we show that platform search creates partially overlapping buyer segments and that list prices determine which segments a property enters. Exploiting list price revisions in a difference-in-differences design, we find that both downward and upward revisions attract buyers from newly reached segments. Downward revisions increase visitor flows persistently, leading to faster sales at lower prices, while upward revisions generate short-lived interest, higher sale probabilities, and higher transaction prices. We develop a search model with buyer-side filtering, segmented consideration sets, and time-consuming inspections that rationalizes these patterns and explains when sellers may benefit from revising their list prices.

Keywords: List prices; Housing markets; Segmentation; Platform economy.
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1 Introduction

For most homeowners, housing constitutes their largest asset. Compared to many other assets, however, houses are highly illiquid. Housing transactions are subject to substantial search frictions: houses are heterogeneous in location, size, and quality; inspections are time-consuming; and at any point in time only a small subset of potential buyers is active in the market. Moreover, houses are largely indivisible and typically financed with debt. As a result, homeowners who intend to sell are exposed to considerable idiosyncratic risk in their financial wealth. In such a frictional environment, it is important for sellers to market their house optimally. Because a house's location is fixed and its observable characteristics are largely invariant in the short run, the list price is the main instrument available to sellers. When a listing attracts insufficient interest, or when inspections do not result in a sale, revising the list price is one of the few margins of adjustment.

The role of list prices has become even more prominent with the rise of online housing platforms. On these platforms, buyers filter listings using observable criteria such as price, floor space, number of rooms, and location. This filtering behavior induces endogenous market segmentation, as heterogeneous buyers sort themselves into partially overlapping submarkets (Piazzesi et al., 2020). In such segmented housing markets, a given list price can place a house inside the consideration set of some buyers while excluding it from others. This mechanism provides a natural explanation for the empirically observed trade-off between sales price and time-on-market (see, e.g., Merlo and Ortalo-Magné (2004)), without appealing to differences in submarket tightness as in directed search models of housing markets (Albrecht et al., 2016), but instead to observed buyer-side filtering on multiple housing characteristics.

In this paper, we study the role of list prices in frictional and segmented housing markets. We develop a housing market model based on the sequential-search framework of Weitzman (1979). Buyers inspect houses sequentially in order of expected utility, which depends on the list price and other ex ante observable characteristics, while learning their idiosyncratic match value only upon inspection. After each inspection, buyers may purchase the current house, continue searching, or return to a previously visited house if it remains available. We incorporate the empirically observed filtering behavior of buyers into this framework by allowing them to search only within preferred ranges of price and housing characteristics, generating partially overlapping buyer segments.

The model delivers two main predictions. First, it generates negative duration dependence without relying on stigma effects. A newly listed house initially has access to a large stock of incumbent buyers already active in the market, some of whom rank the house highly and inspect it shortly after entry. Because inspections are time-consuming, these potential matches are exhausted only gradually. As the listing ages, newly arriving buyers account for an increasing share of potential

demand, producing declining visitor arrivals and sale probabilities with time-on-market. Second, a seller can respond to this declining demand by revising the list price and repositioning the house across buyer segments. A downward revision makes the listing newly searchable to buyers with lower preferred price ranges and also raises its attractiveness among buyers who remain eligible. An upward revision can instead move the listing into the search sets of higher-price buyers, although the increase in attention is expected to be more limited and short-lived because the higher price reduces the listing's attractiveness among existing buyers. The model therefore predicts distinct effects of downward and upward revisions on buyer composition, visits, selling speed, and transaction prices.

We test these predictions using data from a large online housing platform in Beijing. A unique feature of the platform is that, upon registration, all potential buyers must specify preferred ranges for price, floor space, and number of rooms. We first show that these stated preferences strongly predict the characteristics of the houses buyers eventually purchase, confirming that they meaningfully shape search behavior. We then exploit within-listing list price revisions in a difference-in-differences design. Consistent with the segmentation mechanism in the model, price revisions change the composition of visiting buyers. Downward revisions attract buyers with lower preferred price ranges, while upward revisions attract buyers with higher preferred price ranges, even for the same house. In both cases, revisions also generate visits from buyers who had previously ignored the listing. These findings provide direct evidence that list prices act as positioning instruments across buyer segments.[1]

A second distinctive feature of our data is that we observe all visits by registered buyers to listings on the platform, allowing us to test the model's dynamic predictions. Visitor arrival rates and sale probabilities decline sharply with time-on-market, consistent with the model's stock-flow mechanism. Following a downward price revision, the weekly number of visitors increases persistently, sales occur faster, and final transaction prices are lower. Following an upward revision, visitor flows also increase, but the effect is short-lived and dissipates after approximately two weeks. Nevertheless, upward revisions are associated with higher sale probabilities and higher final transaction prices. Because these effects are identified within the same listing, holding housing characteristics fixed, they reflect changes in buyer access and search behavior rather than differences across properties.

Taken together, the theoretical and empirical results provide a unified account of list-price dynamics in segmented housing markets. Search frictions generate declining demand as listings age, while buyer-side filtering gives sellers a means

[1]Upward list price revisions are more common in China than in Western housing markets and constitute a substantial fraction of revisions in our data. While this institutional feature amplifies their empirical relevance, the underlying mechanism—buyer-side filtering in segmented markets—is not specific to this setting.

of renewing demand by repositioning the same house across buyer segments. List prices therefore influence not only the attractiveness of a property, but also which buyers encounter it and when.

Related Literature. Our paper first contributes to the literature on the role of list prices in housing markets. Early work by Horowitz (1992) emphasizes that the list price is a seller-controlled parameter that differs from the seller's reservation price and shapes the distribution of offers. Subsequent work develops richer models of seller pricing dynamics. In particular, Merlo et al. (2015) jointly model initial list prices, price revisions, offer acceptance, and withdrawal decisions, and show that small menu costs can rationalize substantial list-price stickiness. Seller pricing may also reflect behavioral considerations: Andersen et al. (2022) show that reference dependence and loss aversion generate a pronounced kink in listing prices around sellers' original purchase prices.

A large empirical literature documents that list prices affect buyer attention, selling speed, and transaction outcomes. Merlo and Ortalo-Magné (2004) show that higher initial list prices are associated with fewer viewings, higher offers and transaction prices, and longer selling times, while declining visitor flows predict subsequent price revisions. Related studies document links between overpricing, price reductions, buyer arrivals, and time-on-market (Anglin et al., 2003; de Wit and van der Klaauw, 2013; Haurin et al., 2010; Knight, 2002). Guren (2018) further shows that the demand response to list prices is nonlinear, while Koster and Rouwendal (2024) exploit the responses of transaction prices and time-on-market to list prices to infer sellers' discount rates. Together, this literature establishes list prices as an important strategic choice of sellers and documents their effects on buyer attention, liquidity, and transaction outcomes.

A related theoretical literature emphasizes that list prices can shape buyer search itself. Chen and Rosenthal (1996b) and Chen and Rosenthal (1996a) interpret asking prices as ceiling-price commitments that affect buyers' incentives to incur costly search or inspection before learning their idiosyncratic valuations. Directed-search models extend this idea by allowing asking prices to influence where buyers search. Albrecht et al. (2016) show that asking prices can direct buyers toward particular sellers even under limited commitment, while Han and Strange (2016) provide empirical evidence that lower asking prices attract more bidders. More recently, Coles and Smith (2026) connect directed search to stock-flow matching and show how asking-price competition generates different outcomes for new and incumbent listings. Our paper introduces a distinct margin. Because buyers search within explicitly defined price ranges, a list price determines not only the intensity of demand for a property but also which buyers can find and consider it. Price revisions can therefore reposition the same listing across buyer segments, changing both the level and the composition of buyer demand.

Second, the paper relates to the broader literature on search and matching in housing markets. Wheaton (1990) provides an early search-and-matching framework in which costly search and market tightness jointly determine prices and expected time-to-sale. Subsequent work emphasizes how search frictions and market thickness shape housing-market liquidity and dynamics (Head et al., 2014; Ngai and Tenreyro, 2014; Novy-Marx, 2009). A related structural literature models housing search more explicitly. Carrillo (2012) estimates a stationary equilibrium model with simultaneous buyer and seller search, posted prices, bargaining, and endogenous matching, while Fan et al. (2025) develop a dynamic consumer-search model in a non-stationary environment and show how changes in listings and prices affect search duration, visits, and purchase decisions. Empirically, Genesove and Han (2012) document systematic relationships among buyer search duration, the number of homes visited, seller time-on-market, and market tightness. Closely related to our segmentation mechanism, Piazzesi et al. (2020) show that housing search takes place across partially overlapping market segments rather than in a single homogeneous matching market.

Our model differs by opening the buyer's sequential inspection process explicitly. We build on the optimal sequential-search framework of Weitzman (1979), in which consumers rank alternatives according to reservation values and decide sequentially which option to inspect and when to stop searching. Related models of differentiated-product search similarly combine costly search with uncertainty about match quality (Choi et al., 2018; Moraga-González et al., 2022). We adapt this framework to housing, where inspection is costly and reveals match-specific information, and introduce buyer-specific price segments before search begins. As a result, a seller's list price determines both whether a property enters a buyer's searchable set and, conditional on entry, where it is placed in the buyer's inspection order. This allows us to derive seller-level visits and sale probabilities from buyer-level sequential search and to study how price revisions reposition the same listing across segmented demand.

Third, our paper contributes to the growing literature using digital platforms and micro-level search data to study housing markets. Hendel et al. (2009) show that real estate marketing platforms can affect transaction outcomes, while more recent work uses digital trace data to measure housing demand and search behavior directly. van Dijk and Francke (2018) use online listing clicks to measure demand and relate search intensity to market liquidity and prices. Gargano et al. (2023) use individual-level online search histories to study how local house-price experiences affect the breadth and duration of housing search, while Hu et al. (2025) use detailed onsite-visit records to show that more intensive search affects purchase probabilities and transaction prices. A related literature emphasizes the role of real estate agents and intermediaries in shaping housing search, matching, pricing, and market liquidity (Aiello et al., 2026; Gilbukh and Goldsmith-Pinkham,

2024; Hong, 2022). Our data allow us to connect these dimensions of housing search more directly. We observe buyer-level stated search ranges, property visits, listing histories, and eventual transaction outcomes. This allows us to study not only the intensity of search, but also how seller pricing decisions change the composition of buyers who encounter and inspect a listing. In particular, we show that list price revisions reposition the same property across buyer-defined price segments.

The remainder of the paper is organized as follows. Section 2 develops a model. Section 3 introduces the institutional background, data, and stylized facts that motivate the analysis. Section 4 presents the estimation strategy and main empirical results. Section 5 concludes.

2 Model

In this section, we develop a sequential search model that reflects the institutional features of our data and helps interpret the empirical results. Buyers use observable listing characteristics, including price, floor area, and number of rooms, to restrict their searches to particular ranges, generating partially overlapping market segments. Within these segments, buyers follow a Weitzman-style sequential search rule: they rank houses by expected utility, inspect them one at a time at a cost, and decide after each visit whether to stop or continue searching. Because new listings enter while buyers are already active, a newly listed house can immediately attract a stock of incumbent buyers for whom it ranks highly, generating a high initial visitor arrival rate that declines with time-on-market.

This non-stationary demand gives sellers an incentive to revise list prices even in a stationary aggregate market. A price revision repositions the listing across buyer segments: it makes the house newly searchable to some buyers, removes it from the search sets of others, and changes its ranking among buyers who remain eligible. The model therefore predicts that both upward and downward revisions can generate renewed buyer interest, but through different channels and with different dynamic effects on visits, sale probabilities, and transaction prices.

2.1 Environment

We consider a market that consists of a finite number of buyers and sellers (or houses) in discrete time. In our model, sellers are passive. We derive the optimal behavior of buyers and then ask how sellers could benefit from this behavior if they were to maximize the expected present value of the revenue of selling their house.

Buyers maximize the utility that they can derive from buying one of the houses, net of the sales price and the search costs resulting from visits. Indeed, before

buying a house, buyers need to visit a house. They can visit one house per period at a cost $c > 0$. After each visit, the buyer can decide to buy the house, or continue searching. Instead of buying one of the houses on the market, the buyer can also decide to leave the market with his idiosyncratic outside utility u_{i0}.

Following Choi et al. (2018) and Moraga-González et al. (2022), we assume all buyers pay the same price for the same property, but are heterogeneous in their preferences for all other property attributes. Some property attributes can be observed *ex-ante* (online on the platform), while learning about the other attributes requires a visit. The utility function over these house attributes can be written as:

$$u_{ik} = v_{ik} + z_{ik} - p_k,$$

where $v_{ik} = x_k'\beta_i + \zeta_{ik}$. $x_k'\beta_i$ represents the contribution of the observable attributes of house k to the utility of buyer i. ζ_{ik} is an idiosyncratic component of utility known before visiting. The component v_{ik} is thus the consumer's *ex ante* value for the house k, while the component z_{ik} is the matching value that is revealed to the buyer only when she visits seller k and inspects his property.

Houses are horizontally differentiated. Specifically, for each buyer, ζ_{ik} and z_{ik} are independently and identically drawn from the intervals $[\underline{\zeta}, \overline{\zeta}]$ and $[\underline{z}, \overline{z}]$, according to the distribution functions F^{ζ} and F^{z}, respectively. In addition, they are independent of each other. The distribution of v_{ik} is denoted by F^{v}:

$$F^{v}(v \mid x_k, \beta_i) = \Pr\left(\zeta_{ik} \leq v - x_k'\beta_i\right) = F^{\zeta}\left(v - x_k'\beta_i\right), \qquad v \in [\underline{v}, \overline{v}].$$

Given the distribution of v and z, the cumulative distribution function (CDF) of u_{ik} is

$$F^{u}(u) = \Pr(v + z - p_k \leq u) = \int_{\underline{z}}^{\overline{z}} F^{v}(u + p_k - z)\, dF^{z}(z), \qquad u \in [\underline{u}, \overline{u}].$$

2.2 Sequential Buyer Search

In this static version of the model, we abstract from congestion. In particular, for each buyer, all houses that he could potentially buy remain available for recall until he leaves the market. As a result, this static version of the model is essentially a collection of individual buyer decision problems. Moreover, for each buyer, all houses that he could potentially buy are available on the platform when he starts his search. To derive our results, we consider the situation in which at some point in time, one new listing appears, or one house changes its list price. However, we assume that such events are unexpected for buyers, and that buyers will continue to behave as if such events will never happen again.[2]

Because all houses that he could potentially buy are available on the platform when he starts his search, and all houses remain available for recall, the search

[2]This assumption is akin to an 'MIT'-shock in macroeconomics.

problem of the buyer is equal to the sequential search of Weitzman (1979). He shows that the optimal search behavior of buyers is to rank products based on some reservation utility, and inspect them in descending order of this reservation utility. Whenever the highest realized utility after an inspection exceeds the highest remaining reservation utility, the buyer should stop searching and buy the house with the highest realized utility (or leave the market with the outside option).

The reservation utility depends on the price p_k and the *ex-ante* known value v_{ik}, and is thus match-specific. In particular, for each house k and buyer i, it equals

$$r_{ik} = v_{ik} - p_k + z^*,$$

where value z^* is the extra benefit that makes the buyer i indifferent between making an in-person inspection for a house and receiving z^* with certainty. Following Weitzman (1979) and Choi et al. (2018), the boundary value z^* solves

$$\int_{z^*}^{\bar{z}} (1 - F^z(z))\, dz = c.$$

Let us denote the ordering of reservation utilities by

$$r_{i(1)} \geq r_{i(2)} \geq \cdots \geq r_{i(L_i)},$$

where the subscript $i(m)$ denotes the property ranked m-th for buyer i. We denote the length of each buyer's search sequence as L_i. As we will see below, this length is determined by the buyer's outside option u_{i0}: $r_{i(L_i)} \geq u_{i0} > r_{i(L_i+1)}$. Following Ursu et al. (2025), we assume that each buyer in the market makes at least one visit. We have $\Pr(r_{i(1)} \geq u_{i0}) = 1, \forall i$ and $L_i \geq 1, \forall i$.

As explained above, Weitzman (1979) shows that buyers optimally stop searching once the best realized utility exceeds the reservation utility among all remaining houses. Let $\mathcal{N}_i$ be the set of sellers the buyer has visited so far. The stopping rule is thus given by:

$$\max\Big\{u_{i0},\ \max_{j\in\mathcal{N}_i}(u_{ij})\Big\}\ >\ \max_{k\notin\mathcal{N}_i}(r_{ik}).$$

2.3 Visiting Probabilities and Ranking

Given the optimal sequential search behavior of buyers as described by the order and stopping rule described above, we can derive some features of the search process. In particular, the search order and stopping rule together determine the probability that buyer i visits a property k ranked m_{ik} in her search sequence.

To ensure that a listing ranked m_{ik} is reached in a buyer's search process, the buyer must not stop during the inspection of the first $m_{ik} - 1$ listings. Hence, the probability that buyer i visits the m_{ik}-th listing is precisely the probability of continuing search through the first $m_{ik} - 1$ positions. Moreover, since the continuation

probability is multiplicative and strictly less than one at each stage, it follows immediately that the probability of visiting listing k is strictly decreasing in its rank m_{ik}.

Lemma 1 (Visiting Probability and Its Decline with Rank)**.** *Under the Weitzman search order and stopping rule with recall, let k_m denote the house ranked m in buyer i's search order. The probability that buyer i visits the house ranked m is*

$$\underset{ik_m}{\overset{\text{visit}}{\Pr}}(m) = \begin{cases} 1, & \text{if } m = 1, \\ \prod_{s=1}^{m-1} \Pr(u_{ik_s} < r_{ik_m}), & \text{if } m \geq 2. \end{cases}$$

Moreover, the visiting probability is strictly decreasing in rank:

$$\underset{ik_m}{\overset{\text{visit}}{\Pr}}(m) > \underset{ik_{m+1}}{\overset{\text{visit}}{\Pr}}(m+1), \qquad m = 1, \ldots, L_i - 1.$$

Proof. See proof in the Appendix A.1 □

2.4 Segmentation

Now introduce market segmentation as the result of search criteria into the model. Among these criteria, the price range is arguably the most important. For simplicity, we assume that buyers strictly adhere to their stated price range and only visit and buy properties whose list prices fall within that range.

Let $(\underline{p}_i, \overline{p}_i)$ be drawn from a joint distribution $F(\underline{p}, \overline{p})$ supported on $\{\underline{P} \leq \overline{P}\}$. If the list price p_k falls outside this interval, buyer i will neither visit nor consider purchasing the property. Formally,

$$\underset{ik}{\overset{\text{visit,seg}}{\Pr}}(p_k) = \mathbf{1}\{\underline{p}_i \leq p_k \leq \overline{p}_i\} \underset{ik}{\overset{\text{visit}}{\Pr}},$$

and

$$\underset{ik}{\overset{\text{buy,seg}}{\Pr}}(p_k) = \mathbf{1}\{\underline{p}_i \leq p_k \leq \overline{p}_i\} \underset{ik}{\overset{\text{buy}}{\Pr}}.$$

We assume that both price bounds exhibit decreasing densities: the marginal density of the lower bound $f_{\underline{p}}(\underline{p})$ is decreasing in $\underline{p}$, and the marginal density of the upper bound $f_{\overline{p}}(\overline{p})$ is decreasing in $\overline{p}$. This implies that buyers with low price bounds are more common, while those with high price bounds are relatively rare. This assumption is consistent with the empirical distribution of housing prices in the market, where the density declines as prices increase.

This segmentation framework differs from that in Piazzesi et al. (2020) and is more general. Each buyer is allowed to have a personalized market segment, and the same listing may belong to multiple buyers' individual price-based segments.

2.5 Declining Buyer Arrival Rate

Aggregating individual visiting probabilities across the buyer distribution allows us to derive seller-level buyer arrival rates and examine how they evolve with time-on-market. Consider a house k newly listed at $t = 0$, where $t \geq 0$ denotes the number of periods since listing. Buyers begin visiting the house at $t = 1$. At the time of entry, buyer i has already been active for s_i periods and has visited s_i properties, with $s_i \geq 0$.

When house k enters the market, the set of price-eligible buyers is

$$\mathcal{A}_k = \{\, i : \underline{p}_i \leq p_k \leq \overline{p}_i \,\}, \tag{1}$$

with size A_k. Let

$$\mathcal{S}_i = \{\, k : \underline{p}_i \leq p_k \leq \overline{p}_i \,\} \tag{2}$$

denote the set of sellers considered by buyer i, with size S_i. In this static environment, both A_k and S_i are assumed to be constant over time.

A buyer may visit seller k in any period after the listing enters the market. When house k is listed, buyer i assigns it a global rank m_{ik} among all houses, including those already visited. Since buyer i has already inspected s_i houses, seller k is visited in period t if

$$\max\{1,\ m_{ik} - s_i\} = t. \tag{3}$$

If $m_{ik} - s_i < 1$, the new listing ranks ahead of all unvisited incumbent listings and is visited immediately at $t = 1$. Let $\mathcal{I}_k(t) \subseteq \mathcal{A}_k$ denote the set of buyers who visit listing k in period t:

$$\mathcal{I}_k(t) = \{\, i \in \mathcal{A}_k : \max\{1,\ m_{ik} - s_i\} = t \,\}. \tag{4}$$

The probability that buyer i visits seller k in period t can be decomposed into two components: the probability that the listing is reached in period t in the buyer's remaining search sequence, and the probability that the buyer continues searching up to rank m_{ik}:

$$\Pr_{ikt}^{\text{visit}} = \underbrace{\Pr(\max\{1,\ m_{ik} - s_i\} = t)}_{\text{probability of being reached in period } t} \cdot \underbrace{\Pr_{ik}^{\text{visit}}(m_{ik} \mid \max\{1,\ m_{ik} - s_i\} = t)}_{\text{probability of continuing to rank } m_{ik}}. \tag{5}$$

Aggregating across buyers yields the expected number of visits to seller k in period t:

$$\begin{aligned} V_{kt} &= \sum_{i \in \mathcal{I}_k(t)} \Pr_{ikt}^{\text{visit}} \\ &= \sum_{i \in \mathcal{A}_k} \Pr(\max\{1,\ m_{ik} - s_i\} = t)\ \Pr_{ik}^{\text{visit}}(m_{ik} \mid \max\{1,\ m_{ik} - s_i\} = t). \end{aligned} \tag{6}$$

Lemma 1 shows that the individual visiting probability $\Pr_{ik}^{\text{visit}}(m_{ik})$ is decreasing in the rank m_{ik}. We next characterize the ranking probability. When a new

listing appears, buyer i draws its reservation utility from the CDF F^r and inserts it into the search sequence. Because the reservation utilities of incumbent and newly listed houses are independently drawn from the same continuous distribution, the new listing is ex ante equally likely to occupy any of the $S_i + 1$ possible global ranks. Given that buyer i has already visited s_i houses, the new listing is visited immediately if it falls within the first $s_i + 1$ positions. Therefore,

$$\Pr(\max\{1,\ m_{ik} - s_i\} = t \mid S_i) = \begin{cases} \dfrac{s_i + 1}{S_i + 1}, & \text{if } t = 1, \\ \dfrac{1}{S_i + 1}, & \text{if } t = 2, \ldots, S_i + 1 - s_i, \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

This expression implies that a newly listed house is disproportionately likely to appear in the first position of the buyer's remaining search sequence. Intuitively, the buyer has already visited s_i relatively attractive houses under the optimal search order, so the remaining incumbent listings are drawn from a lower, truncated portion of the reservation-utility distribution. By contrast, the new listing is drawn from the full distribution and therefore has a higher probability of ranking ahead of the remaining options. The first-period probability is therefore elevated to $(s_i + 1)/(S_i + 1)$, while each later feasible period receives mass $1/(S_i + 1)$.[3]

Because V_{kt} is the sum of individual visiting probabilities and the two components in equation (6) enter multiplicatively, the time profile of visits reflects both components. The first component places disproportionately large mass on $t = 1$, generating an initial spike in visits. The second component, $\Pr_{ik}^{\text{visit}}(m_{ik})$, is strictly decreasing in m_{ik}. Together, these forces imply that visits are highest immediately after listing and decline with time-on-market, provided the cross-sectional distribution of (S_i, s_i) is stable over time.

Proposition 1 (Decreasing buyer arrival rate)**.** *When the number of buyers in $\mathcal{A}_k$ and the number of sellers in $\mathcal{S}_i$ remain constant over time, the number of visits a seller receives declines with the seller's time-on-market t. When a new listing appears, it is disproportionately likely to be placed first in buyers' remaining search sequences because the remaining incumbent listings are drawn from a truncated distribution. Consequently, the expected buyer arrival rate V_{kt} is exceptionally high in the first period and strictly decreases thereafter.*

This proposition implies that, under buyers' optimal sequential search behavior, the model naturally predicts a declining buyer arrival rate as time-on-market

[3] If no incumbent buyers are present when the seller enters the market and all buyers enter at $t = 0$, then $s_i = 0$. In this case,

$$\Pr(\max\{1,\ m_{ik} - s_i\} = t \mid S_i) = \frac{1}{S_i + 1}$$

for all feasible $t \geq 1$. Thus the buyer's global rank of seller k, m_{ik}, coincides with the visiting period t, and the ranking probability is constant across periods.

increases. This prediction is consistent with the stylized fact in Figure 2: buyer arrival rates fall sharply at the beginning and then flatten out over time.

2.6 List Price Change

To explain our empirical findings that list price revisions attract buyers from different price segments, generate more visits, and increase sale probabilities, we examine the effect of list price adjustments within the sequential search framework. We focus on a single seller who changes the list price while holding all other conditions fixed, and show how the price-segmentation mechanism and the declining buyer arrival rate jointly shape the response of buyer visits and transaction probabilities.

Visitor Composition. Recall that the market is segmented by list prices and buyers' stated price ranges. For seller k, the set of price-eligible buyers is

$$\mathcal{A}_k = \{\, i : \underline{p}_i \le p_k \le \overline{p}_i \,\},$$

which includes all buyers whose admissible price range contains the list price p_k. A change in p_k therefore changes the composition of $\mathcal{A}_k$. When the list price increases, buyers with lower admissible price ranges exit this set, and the distribution of admissible intervals $[\underline{p}_i, \overline{p}_i]$ shifts upward. Conversely, when the list price decreases, buyers with lower admissible price ranges enter $\mathcal{A}_k$, shifting the distribution downward. Assuming that $(\underline{p}_i, \overline{p}_i)$ is drawn from a continuous distribution with positive density, even a small price adjustment induces some buyers to enter and others to exit $\mathcal{A}_k$.

Lemma 2 (List price changes and buyer composition)**.** *Let*

$$\mathcal{A}_k(p) = \{\, i : \underline{p}_i \le p \le \overline{p}_i \,\}$$

denote the set of buyers for whom seller k's list price p falls within their admissible price interval $[\underline{p}_i, \overline{p}_i]$. Suppose that $(\underline{p}_i, \overline{p}_i)$ are drawn from a joint continuous distribution with positive density over the relevant range.

Then a nonzero change in the list price changes the composition of price-eligible buyers. In particular, an increase in the list price shifts the admissible price ranges of eligible buyers upward:

$$p'_k > p_k \quad \Rightarrow \quad \begin{cases} \mathbb{E}[\underline{p}_i \mid i \in \mathcal{A}_k(p'_k)] > \mathbb{E}[\underline{p}_i \mid i \in \mathcal{A}_k(p_k)], \\ \mathbb{E}[\overline{p}_i \mid i \in \mathcal{A}_k(p'_k)] > \mathbb{E}[\overline{p}_i \mid i \in \mathcal{A}_k(p_k)]. \end{cases}$$

Conversely, a decrease in the list price shifts these admissible price ranges downward.

Lemma 2 shows that raising the list price shifts the admissible price intervals of eligible buyers upward, while lowering it shifts them downward. This provides

a theoretical rationale for the empirical findings in Section 4.2. Under price-based market segmentation, a list price revision moves the property across buyer price ranges and therefore attracts buyers with different price preferences. This explains why, after a price revision, both the average upper and lower bounds of visiting buyers' preferred price ranges move in the same direction as the list price.

Buyer arrival. Lemma 2 implies that, when the seller changes the list price p_k, the arrival flow $V_{kt}(p_k)$ is affected through two channels. First, the set of price-eligible buyers $\mathcal{A}_k = \{\, i : \underline{p}_i \leq p_k \leq \overline{p}_i \,\}$ changes. After adjusting prices, the seller lost some potential buyers but entered the preferred range of some new buyers. Second, for buyers who remain in $\mathcal{A}_k$, the ranking m_{ik} changes because the relative attractiveness of the listing varies with price. The conditional visiting probability $\Pr^{\text{visit}}_{ikt}(m_{ik})$ thus changed. For example, an increase in price causes a decrease in reservation utility r_{ik}, which in turn lowers the ranking. Following the logic from Lemma 1, $\Pr^{visit}_{ikt}$ will then decrease for any period t.

We consider the price change from p to p' at time period $t > 1$. The change of number of visits can be expressed by:

$$\Delta V_{kt} = V_{kt}(p') - V_{kt}(p)$$

$$= \underbrace{\sum_{i \in \mathcal{A}_k(p') \setminus \mathcal{A}_k(p)} \overset{\text{visit}}{\underset{ik,1}{\Pr}}(p')}_{\text{(i) entry of new buyers}} - \underbrace{\sum_{i \in \mathcal{A}_k(p) \setminus \mathcal{A}_k(p')} \overset{\text{visit}}{\underset{ikt}{\Pr}}(p)}_{\text{(ii) exit of former buyers}} + \underbrace{\sum_{i \in \mathcal{A}_k(p') \cap \mathcal{A}_k(p)} \left[\overset{\text{visit}}{\underset{ikt}{\Pr}}(p') - \overset{\text{visit}}{\underset{ikt}{\Pr}}(p) \right]}_{\text{(iii) pure price effect for remaining buyers}}.$$

The first term captures buyers who become price-eligible under the new price p'. For these buyers, the listing was previously outside their admissible price range and enters their search set as if newly listed. Their relevant visiting probability is therefore evaluated at period $t = 1$, where t denotes time since the listing becomes searchable for them rather than calendar time. The second term captures buyers who become ineligible after the price change and no longer consider the listing. Their last relevant visiting probability is evaluated at the seller's current time-on-market t. The third term captures buyers who remain eligible under both prices; for them, the price change affects visits only through the induced change in ranking and continuation probabilities.

Consider first a price reduction, $p' < p$. The net entry-exit effect is positive. By Proposition 1, newly eligible buyers enter at period $t = 1$, when visiting probabilities are highest, whereas exiting buyers are evaluated at the current time-on-market t, when visiting probabilities are lower. Moreover, if buyers' price bounds have decreasing density, a price reduction brings in more newly eligible buyers than it removes. Among buyers who remain eligible, the lower price further increases the probability of visiting. Hence, a price reduction unambiguously increases the seller's total visits.

Although the effect of a price increase is theoretically ambiguous, it can be positive when the new-buyer effect dominates. A higher price moves the listing into a higher price segment and admits a new set of buyers. These newly eligible buyers treat the listing as newly searchable and therefore evaluate it at period $t = 1$, when visiting probabilities are highest. This high-weight inflow may offset, or even dominate, the loss of buyers who exit the admissible price range. The offsetting force among buyers who remain price-eligible is likely to be more limited: although the higher price reduces the listing's attractiveness and lowers their visiting probability, the listing still remains within their admissible price range. Therefore, the net effect of a price increase can be positive, but if so, it is likely to be short-lived because visiting probabilities decline sharply with time-on-market. We summarize these implications in the following proposition.

Proposition 2 (Effects of list price change on buyer arrival)**.** *Consider a change in the list price from p to p' at time period $t > 1$, and let $\mathcal{A}_k(p_k)$ denote the set of price-eligible buyers. Then the change in the number of visits to seller k in period t can be decomposed into (i) the entry of newly eligible buyers, (ii) the exit of formerly eligible buyers, and (iii) the pure price effect on buyers who remain eligible.*

(a) ***Price decrease ($p' < p$).*** *By Proposition 1, the number of newly entering buyers always exceeds the number of exiting buyers, and remaining buyers experience a higher visiting probability. Hence, a price decrease unambiguously increases the seller's total number of visits.*

(b) ***Price increase ($p' > p$).*** *A higher list price admits a new set of buyers, but under the decreasing–density assumption, the number of new entrants may be slightly smaller than the number of exiting buyers. Nevertheless, all newly entering buyers place the seller in period $t = 1$, where the visiting probability is highest, so their contribution to the total arrival rate can exceed that of the buyers who exit. Among buyers who remain eligible under the higher price, the visiting probability decreases, but this negative effect is comparatively small. Taken together, the net impact of a price increase is theoretically ambiguous, but it is likely to be positive in the short run. As the listing stays longer on the market and visiting probabilities decline sharply with t, the long-run effect of a higher price may become negative.*

Proposition 2 provides a theoretical rationale for the empirical findings in Section 4.3. Under the assumption of market segmentation and the proposition of decline buyer arrival rate in Proposition 1, a price decrease unambiguously increases visits, whereas a price increase does not necessarily reduce traffic and may even lead to more visits. This proposition helps explain why, in our empirical results , an upward revision in the list price can also lead to more visits and at the same time an downward revision leads to more visits.

3 Background, Data, and Stylized Facts

3.1 Background and Data Description

We use a unique dataset from the largest real estate agency in China. The agency has a nationwide market share of 26.6% and a second-hand property transaction volume of 1.94 trillion Chinese yuan by the end of 2020. Established in Beijing, it holds more than 50% of the city's second-hand housing market, making the dataset highly representative of Beijing's housing transactions. The data cover second-hand housing transactions, buyer visits, and list prices for properties offered for sale in Beijing from August 2015 to July 2016.

Buyers can access the platform to get the information about listed properties and browse the website to identify suitable houses. They can also subscribe to properties they are interested in and receive notifications when listing information changes. They must register before arranging offline property visits through brokers. Upon registration, buyers report their preferences regarding price range, square meters, and number of rooms.

Sellers similarly register on the platform to list their properties for sale. Housing transactions are intermediated by real estate brokers, who facilitate matching between buyers and sellers and coordinate the transaction process. Listing brokers advise sellers on pricing, marketing, and negotiations, while buyer brokers assist buyers in identifying suitable properties, arranging visits, evaluating alternatives, and negotiating purchases. Brokers therefore play both informational and advisory roles, helping market participants navigate search frictions and transaction complexity. Property visits must be arranged through brokers, who also serve as the primary channel of information exchange between buyers and sellers.[4]

After a property is listed, sellers may update relevant listing information, including the list price. To revise the price, a seller must inform the broker, who then updates the information on the platform. The full history of list price revisions is publicly visible on the website, allowing buyers to observe both the timing of each adjustment and the list price before the adjustment. In addition, price revisions may reach buyers through two notification channels. First, buyers who have subscribed to a specific property automatically receive an alert when its list price is revised, regardless of whether the revision is upward or downward. Second, brokers may directly inform buyers and invite them to visit the property when a revised list price matches their stated preferences.

However, apart from such notifications, price-revised properties are not mechanically prioritized or promoted in the ordering of listings displayed on the website. The platform may use recommendation algorithms to rank certain listings more prominently when they appear to offer better value for money. Yet list-

[4]Figure A1 illustrates the operation of the platform and the matching process between buyers and sellers.

ings are not promoted simply because they have recently adjusted their prices. If anything, a price reduction may make a listing more likely to be recommended by improving its perceived price-quality ratio, although this need not occur for every markdown. By contrast, a price increase should, in principle, make such recommendation less likely, as it worsens the listing's perceived price-quality ratio.

This dataset contains five sub-datasets which can be connected via the house and buyer identifiers: transaction records, visit records (which buyer visits which house), property characteristics, buyer characteristics (that is, the housing preferences referred to above as well as gender), and list price adjustment records. Our data records more than 4 million visits and 0.7 million price revisions, which allows us to investigate the impact of the list price on the sales process by exploiting list price variation over time, while controlling for seller/house characteristics through fixed effects. Our dataset has many similarities to that of Merlo and Ortalo-Magné (2004) and Merlo et al. (2015), who also have detailed information about the matching process, but we have a much larger number of observations and information about buyer preferences.

3.2 Summary Statistics

In this section, we report descriptive statistics for the sample. We begin by reporting descriptive statistics on housing characteristics and list price adjustment behavior. Our research design relies on a selected sample that includes properties with either exactly one list price revision or no list price revision during the listing period. Table 1 presents number and magnitude of list price revisions for both the full sample and selected sample. Our selected sample includes 191,219 properties, of which 18.76 percent experience an upward price revision and 18.69 percent experience a downward price revision. The remaining properties constitute the control group. Table A1 reports descriptive statistics for price revisions and housing characteristics in both the full sample and the selected sample. Housing characteristics are highly similar across the two samples, with no statistically significant differences. This evidence suggests that, although the baseline analysis relies on a subset of the data, the selected sample remains broadly representative of the full sample.

Our data also record buyers' stated preferences over price, housing size, and the number of bedrooms. Table A2 presents summary statistics on buyers' stated search criteria and visiting behavior for buyers who made at least one property visit. On average, buyers state a preferred price range from 3.3 to 4.1 million Chinese Yuan, a preferred floor area range from 77 to 100 square meters, and a preferred bedroom range from 1.9 to 2.4 bedrooms. The average buyer makes 5.8 property visits, with substantial dispersion across buyers.

Using visit records in the selected sample, we aggregate visiting buyers' stated preferences to the property-week level. Weekly visit counts are constructed for all

Table 1: Summary Statistics: List Price Revisions

	Full sample		Selected sample	
	Upward	Downward	Upward	Downward
Number of revisions	229,685	322,657	35,874	35,731
Number of properties	140,911	167,180	35,874	35,731
Proportion of properties	35.85%	42.54%	18.76%	18.69%
Average number of revisions	1.63	1.93	1.00	1.00
Average revision magnitude	4.74%	-3.13%	4.62%	-3.20%

Notes: This table reports the number and magnitude of list price revisions among revised properties in the full sample and the selected sample, separately for upward and downward revisions. Number of properties refers to properties with upward or downward list price revisions. In the full sample, upward and downward revisions may overlap, whereas they are mutually exclusive in the selected sample. Average number of revisions is calculated among revised properties. Average revision magnitude is the average percentage change in the list price for each revision.

property-week observations, with weeks without visits assigned a value of zero. This yields two property-week panels: a markup sample of about 524,000 observations and a markdown sample of about 536,000 observations. Each panel includes properties with one price revision and properties with no price revision. The average number of weekly visits is 1.9. Buyer's upper and lower price bound, however, are averaged only over property-week observations with at least one visit, as they are undefined when no buyers visit. We get a markup sample of about 379,000 observations and a markdown sample of about 393,000 observations. Because the panel is highly unbalanced, each property has at least one recorded visit in only 3.5 weeks on average during the sample period.

3.3 Stylized Facts

3.3.1 Considerable Share of Upward Revisions

One distinctive feature of our data is that it records a substantial number of list price revisions, including many upward revisions, which are relatively uncommon in other housing markets. Merlo et al. (2015) document that 77% of sellers in England never change their list price, a pattern they attribute to fixed menu costs. Similarly, de Wit and van der Klaauw (2013) show that 80% of sellers in the Dutch housing market never change their list price, and that upward price adjustments are very rare in their data.

By contrast, list price revisions are frequent in China's housing market. In our full sample, about 60% of properties experience at least one price revision during the listing period. As shown in Table 1, properties with upward revisions ac-

count for approximately 36% of all properties and 46% of all revised properties in the full sample, indicating that upward revisions constitute a substantial share of price adjustments. At the same time, upward revisions do not imply that list prices increase on average over the listing spell. Despite the prevalence of upward revisions, Figure A4 shows that the ratio of the current list price to the initial list price declines gradually with time on market, indicating an overall downward trend in list prices.

The housing market boom in China during 2015–2016 provides a potential explanation for the prevalence of upward revisions. As shown in Figure A5, both list prices and transaction prices increased sharply over our one-year sample period. Initial list prices rose by about 30%, while sale prices increased by about 20%. In such a rapidly rising market, sellers may adjust their expectations upward and revise list prices in response to changing market conditions.

Table 1 provides additional evidence on list price adjustment behavior. In the full sample, sellers with upward revisions increase their list prices 1.63 times on average, while sellers with downward revisions decrease their list prices 1.93 times on average. The average upward revision is 4.74%, compared with an average downward revision of 3.13%. In the selected sample, approximately 38% of properties experience exactly one revision, split almost evenly between upward and downward revisions. Figure A3 presents the distribution of list price changes in the selected sample. Overall, list price revisions are relatively common in China, and upward revisions account for a substantial share of all revisions, although they remain less frequent than downward revisions.

3.3.2 The Importance of Search Criteria

When searching for housing in the market, buyers are typically not searching at random. Instead, they have preferences over attributes such as location, quality, and price, and tend to focus their search within specific market segments defined by these preferences. A unique feature of our data is that it records buyers' stated preferences for housing price, floor space, and number of bedrooms at the time of registration on the platform. Although these stated search ranges are often relatively broad, and we do not observe whether buyers subsequently adjust their search criteria during the search process, these preferences are nevertheless important, as they define segments of the market.

Our data show that buyers use these stated search ranges to direct their search, although adherence is not perfect. Figure 1 illustrates this finding using price search ranges as an example. The figure shows a high degree of consistency between the average list prices of the houses buyers visited and their stated lower and upper price bounds. The list price falls strictly within buyers' average stated price ranges in smaller-size market segments, while in larger-size segments it lies slightly above buyers' average stated price ranges.

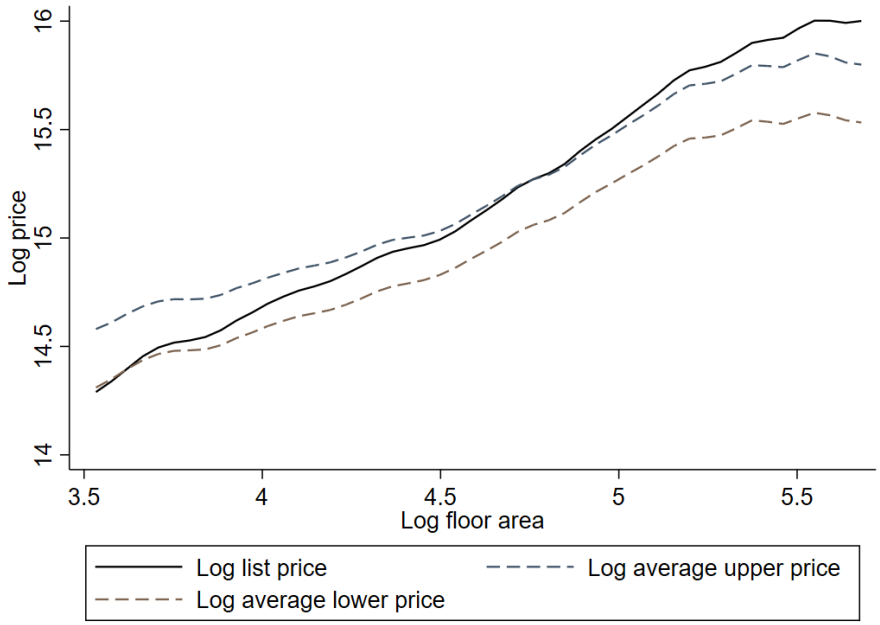


Figure 1: List Prices and Buyer Price Preferences over Floor Area

Notes: The horizontal axis reports the logarithm of floor area, and the vertical axis reports log prices. The three lines plot the property's actual list price, the average upper price bound among all visitors, and the average lower price bound among all visiting buyers.

Search criteria are also crucial for predicting the characteristics of properties that buyers ultimately purchase. Table 2 reports regressions of the transaction price, floor space, and number of bedrooms of the purchased property on the corresponding upper and lower bounds of buyers' stated preference ranges. The results show that buyers' stated preferences are strong predictors of realized purchase outcomes. Stated price preferences explain more than 80% of the variation in transaction prices, stated size preferences explain about 70% of the variation in purchased size, and stated bedroom preferences explain roughly 50% of the variation in the number of bedrooms of purchased properties. The sums of the coefficients on the upper and lower bounds are 0.93, 0.91, and 0.81, respectively. These estimates indicate that upward shifts in stated preference ranges are strongly associated with higher realized property characteristics, with the first two dimensions exhibiting an approximately proportional relationship. Notably, these explanatory powers are achieved even without additional control variables or fixed effects.

These results suggest that buyers' stated search criteria are credible, particularly with respect to price ranges. Buyers appear to adhere closely to their pre-stated search ranges when searching for and purchasing houses.

3.3.3 Duration Dependence

Our data also reveals two key patterns in buyer arrivals and sale outcomes over a property's time-on-market.

Table 2: Buyers' Search Criteria and Characteristics of Purchased Houses

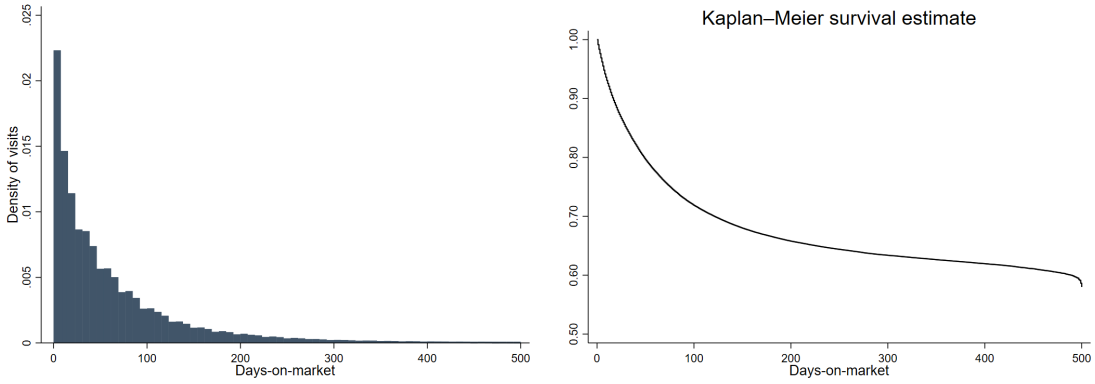
	log(purchase price)	log(purchase size)	log(purchase bedrooms)
	(1)	(2)	(3)
log(upper price)	0.673***		
	(0.004)		
log(lower price)	0.257***		
	(0.005)		
log(upper size)		0.468***	
		(0.004)	
log(lower size)		0.442***	
		(0.004)	
log(upper bedrooms)			0.382***
			(0.003)
log(lower bedrooms)			0.425***
			(0.003)
N	180,029	181,074	181,813
R^2	0.839	0.707	0.503

Notes: This table reports regressions of the characteristics of the property purchased by the buyer on the upper and lower bounds of the buyer's stated preferences reported at registration. The dependent variables are the log purchase price, log purchase size, and log number of bedrooms of the purchased property. Standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

First, buyer arrivals are heavily concentrated early in the listing period. Panel (a) of Figure 2 presents the distribution of visits over time-on-market. The figure shows that most visits occur shortly after listing, with the density declining rapidly as time-on-market increases. This indicates that buyer arrival rates are strongly front-loaded. Second, a similar pattern emerges for sale outcomes. Panel (b) of Figure 2 plots the cumulative probability of sale over time-on-market using a Kaplan–Meier survival estimate. The survival curve becomes progressively flatter, indicating that the hazard of sale declines with time-on-market. In other words, properties are most likely to sell early in the listing period, with the likelihood of transaction decreasing as listings age.

Taken together, these patterns suggest that both buyer arrivals and sale hazards are strongly duration dependent over the listing lifecycle, with the highest activity occurring shortly after listing and declining substantially thereafter. The similarity in these patterns further points to a close relationship between buyer arrival dynamics and transaction outcomes. These patterns have been explained using a stock–flow matching framework in housing markets (Gilbukh, 2023; Smith

et al., 2026). In a stock–flow environment, newly listed properties are immediately exposed to the entire stock of active buyers, generating a high initial arrival rate of potential matches. As time passes, the stock of buyers who find the property acceptable is gradually exhausted, leaving only the flow of newly arriving buyers. Consequently, both the buyer arrival rate and the probability of sale decline with time-on-market. Our empirical findings are consistent with this prediction: listings experience the highest visiting traffic and sale probabilities shortly after entering the market, followed by a steady decline over time.



(a) Visits over time-on-market (b) Sale hazard over time-on-market

Figure 2: Buyer Arrivals and Sale Probabilities over Time-on-market

Notes: Panel (a) shows the distribution of all visits over sellers' time-on-market. Panel (b) presents Kaplan–Meier survival estimates for property listings, where a sale is defined as the failure event. All properties enter the risk set at the time of listing and are right-censored at the end of August 2016, allowing us to characterize the time-to-sale process and the corresponding sale hazard.

3.3.4 List Price Revisions and Sale Outcomes

Sale prices. To examine whether price revisions are associated with final transaction prices, we regress the sales price on the number of upward revisions and the number of downward revisions, controlling for the initial list price and other housing characteristics. The results presented in Table 3 show that the number of revisions during the listing period is highly correlated with the final sales price. On average, each additional upward revision is associated with a sales price that is about 4% higher, while each additional downward revision is associated with a sales price that is about 2.4% lower. In addition, the initial list price is also significantly positively associated with the final transaction price. More frequent revisions are generally associated with a longer list duration. As can be seen in columns (3) and (4), this effect is robust to controlling for time-on-market, measured as the natural logarithm of the number of days between the initial listing date and the transaction date. Although these estimates should not be interpreted causally, they indicate that, beyond the initial list price, subsequent price revisions during the selling process are associated with the final transaction price.

The results are consistent with the standard view in the literature that list prices are positively associated with, and may have a positive causal effect on, transaction prices (Guren, 2018; Knight, 2002; Koster and Rouwendal, 2024; Merlo and Ortalo-Magné, 2004). Going beyond this literature, we show that not only is the initial list price significantly positively associated with the transaction price, but subsequent list price revisions are also strongly related to final sale prices. Upward revisions are associated with higher transaction prices, while downward revisions are associated with lower transaction prices.

Table 3: List Price Revisions and Sale Prices

	log(sale price)			
	(1)	(2)	(3)	(4)
Number of upward revisions	0.039***	0.039***	0.038***	0.038***
	(0.000)	(0.000)	(0.000)	(0.000)
Number of downward revisions	-0.023***	-0.023***	-0.024***	-0.024***
	(0.000)	(0.000)	(0.000)	(0.000)
log(list price)	0.947***	0.947***	0.944***	0.944***
	(0.002)	(0.002)	(0.002)	(0.002)
log(listing duration)			0.002***	0.003***
			(0.000)	(0.000)
N	156,864	156,369	156,860	156,365
R^2	0.994	0.994	0.994	0.994
House attributes	✓	✓	✓	✓
Complex F.E.	✓	✓	✓	✓
Month F.E.	✓	✓	✓	✓
Neighborhood × Month F.E.		✓		✓

Notes: The dependent variable is the logarithm of the transaction price of the property. The main explanatory variables are the total numbers of upward and downward list price revisions during the listing period. House attributes include property characteristics such as size, age, floor level, and housing type. All specifications include apartment-complex fixed effects and transaction month fixed effects. Columns (2) and (4) further control for neighborhood-by-month fixed effects to account for time-varying local housing market conditions. Standard errors are clustered at the apartment-complex level. *** p<0.01, ** p<0.05, * p<0.1.

Sale probability. To investigate the impact of list price change on sale probability, we estimate a duration model similar to de Wit and van der Klaauw (2013). The model is estimated using an extended Cox proportional hazard specification on a

house-week panel. The dependent variable is the sale hazard, defined as the conditional probability that property i is sold at duration t, given that it has remained on the market until t. The baseline hazard consists of three components. First, $\lambda(t)$ captures duration dependence in time-on-market. Second, $\psi_n(n_i)$ controls for neighborhood-specific market conditions. Third, $\psi_w(w_t)$ captures aggregate calendar-week effects.

The key explanatory variable is the cumulative number of price revisions up to duration t, denoted by R_{it}. This variable is time-varying and increases whenever the seller revises the list price. The coefficient δ therefore measures how cumulative price revisions shift the sale hazard conditional on the property remaining unsold up to duration t. The vector X_i includes additional property characteristics and seller controls.

$$\theta_i(t \mid X_i, n_i, w_t, R_{it}) = \lambda(t)\psi_n(n_i)\psi_w(w_t)\exp\left[X_i'\gamma + \delta R_{it}\right]. \tag{8}$$

Table 4 shows that price revisions are strongly associated with higher sale hazards. Columns (1) and (2) indicate that both cumulative upward and downward revisions increase the probability of sale, conditional on remaining on the market, with downward revisions having a larger effect. Columns (3) and (4), which restrict the sample to properties with either exactly one revision or no revision, show a similar pattern: relative to properties without price changes, a single upward revision increases the purchase hazard by about 70%, while a single downward revision more than doubles it.

These results confirm that price cuts substantially accelerate transactions, consistent with the standard view that lower list prices improve sale hazards(de Wit and van der Klaauw, 2013). The standard view is that higher list prices are typically associated with longer time on market or a lower probability of sale (Knight, 2002; Koster and Rouwendal, 2024; Merlo and Ortalo-Magné, 2004). More surprisingly, we found that upward revisions are positively associated with sale hazards. Figure 3 presents the dynamic effects of price revisions on sale hazards. Both upward and downward revisions generate a sharp increase in the hazard immediately after the revision, with larger effects for markdowns. The effects gradually decay over time but remain above one for several weeks. Overall, the dynamic evidence reinforces the Table 4: both markdowns and markups are associated with higher sale hazards after the revision, with stronger and more persistent effects for markdowns.

The results reveal a puzzle in the relationship between list price revisions and transaction outcomes. While downward price revisions substantially increase the probability of sale, consistent with the standard negative relationship between price and demand, upward price revisions are also associated with significantly higher sale hazards. This finding is difficult to reconcile with a conventional static demand framework and suggests that list price revisions may affect buyer behavior through channels beyond the direct price effect.

Table 4: List Price Revisions and Sale Hazard

	Sale hazard (Cox model)			
	All revisions		One or zero revision	
	(1)	(2)	(3)	(4)
Cumulative upward revisions	1.370***	1.397***		
	(0.008)	(0.008)		
Cumulative downward revisions	1.688***	1.645***		
	(0.008)	(0.008)		
Upward revision			1.707***	
			(0.023)	
Downward revision				2.369***
				(0.028)
log(list price)		0.495***	0.683***	0.863***
		(0.020)	(0.029)	(0.037)
N	6,772,149	5,817,890	2,619,063	2,571,019
House attributes		✓	✓	✓
Neighborhood strata	✓	✓	✓	✓
Week strata	✓	✓	✓	✓

Notes: This table reports hazard ratios from Cox proportional hazard models estimated on a house-week panel. The dependent variable is the purchase hazard. Columns (1) and (2) examine the relationship between cumulative price revisions and sale hazards. Columns (3) and (4) estimate the treatment effect of a single price revision on the purchase hazard. The sample in columns (3) and (4) is restricted to properties that experience exactly one price revision or no price revision during the listing period. Column (3) compares properties with exactly one upward price revision to properties with no price revisions, while column (4) compares properties with exactly one downward price revision to properties with no price revisions. All specifications allow the baseline hazard to vary across neighborhoods and calendar weeks through stratification. Since the observation window ends in July 2016, the survival duration for each house-week observation is right-censored at July 31, 2016. Houses that had not been sold by this cutoff date are treated as censored observations. Standard errors are clustered at the apartment-complex level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

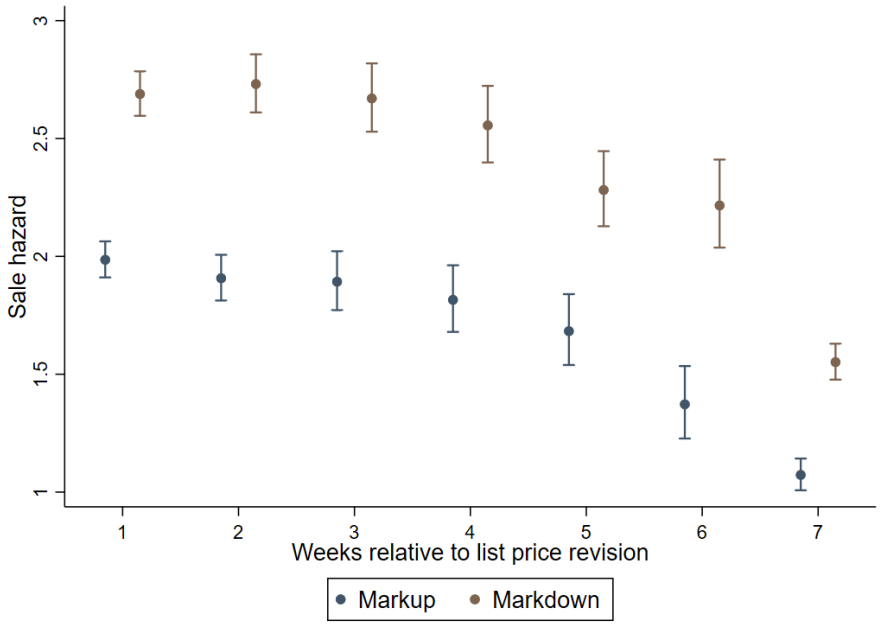


Figure 3: Dynamic Effects of List Price Revisions on Sale Hazards

Notes: This figure reports the dynamic effects of a seller's price revision on the sale hazard over periods 1 to 7 after the revision, estimated from the Cox model specified in Equation (8).

4 List Price Revisions and Visits

To investigate the mechanisms underlying the puzzle, the following empirical analysis examines how list price revisions reshape buyer composition and search activity on the platform. Specifically, we employ a staggered difference-in-differences design to study how price revisions affect both the composition of buyers visiting a property and the intensity of visits over time. We first examine whether price revisions change the average upper and lower bounds of the price ranges indicated by visiting buyers. We then analyze how list price revisions affect the weekly number of visits received by a property. Finally, we study whether price revisions attract new visitors whose initially stated price ranges did not include the property at its original list price. Together, these analyses shed light on how seller pricing behavior interacts with heterogeneity in buyer search behavior and attention allocation.

4.1 Empirical Strategy

4.1.1 Identification

A central empirical challenge is that list price revisions are not randomly assigned. Both the decision to revise the list price and the timing of the revision are chosen by sellers and may reflect information about the property, the seller's reservation value, or contemporaneous market conditions. In our data, Figure A4 documents that, although all sellers tend to lower their list prices as time-on-market increases,

sellers in higher-priced market segments are more likely to raise their list prices, whereas sellers in lower-priced segments are more likely to cut their list prices. This pattern generates divergence in market prices across properties in different price ranges.

Our staggered difference-in-differences design addresses these concerns by comparing changes in outcomes around the timing of a price revision for treated properties with contemporaneous changes for properties whose list prices remain unchanged. The key source of identifying variation comes from the timing of a single upward or downward revision among otherwise comparable listings. To make the treated and control properties comparable, the baseline specification includes highly granular local area-by-week fixed effects. In particular, we control for complex-by-week fixed effects, and in alternative specifications use neighborhood-by-week fixed effects. A residential complex typically consists of multiple highly similar buildings developed by the same developer, with almost the same location, building age, layout, amenities, school access, and surrounding environment. As a result, complex-by-week fixed effects absorb highly localized and time-varying demand shocks, such as changes in buyer traffic, neighborhood sentiment, seasonal market fluctuations, and other common shocks affecting similar properties in the same local market at the same time.

Conditional on these granular local area-by-week fixed effects, the comparison is therefore made among highly similar properties listed in the same local market and observed in the same calendar week. The remaining variation comes from idiosyncratic differences in sellers' price-adjustment decisions, such as seller-specific beliefs, bargaining strategies, urgency to sell, or private reservation values, rather than from broad local market movements. This specification substantially reduces the concern that the estimated effects are driven by differential trends across locations or by common demand shocks that simultaneously affect both list price revisions and buyer behavior.

The causal interpretation of the estimates relies on the standard parallel-trends assumption. Specifically, the assumption is that absent the price revision, treated properties would have experienced the same evolution in buyer composition and search activity as comparable unrevised properties within the same local market and week. This assumption does not require price revisions to be randomly assigned unconditionally. Rather, it requires that, after conditioning on property characteristics, listing fixed effects, and granular local area-by-week shocks, the timing of a single price revision is not correlated with unobserved shocks that would have independently changed the outcomes at the same time. We assess the plausibility of this assumption using event-study specifications that examine whether treated and control properties exhibit similar outcome trends before the revision. The absence of differential pre-trends would support the identifying assumption, while post-revision changes can then be interpreted as the effect of the

list price revision on buyer composition and search activity.

4.1.2 Staggered DID Specification

We employ a staggered difference-in-differences design to study how list price revisions affect buyer composition and search activity. Properties without any price revisions serve as the control group, while properties experiencing exactly one price revision constitute the treatment group. For upward revisions, the treatment group consists of properties with a single markup, while the control group contains properties whose list prices remain unchanged throughout the listing period. Similarly, for downward revisions, the treatment group includes properties with a single markdown, and the control group again consists of properties without price revisions.[5]

We begin with a canonical two-way fixed effects specification estimated at the property-week level:

$$Y_{it} = \alpha + \beta\{Up_{it}, Down_{it}\} + \lambda X_{it} + \gamma_i + \phi_t + \mu_{it}. \tag{9}$$

where Y_{it} denotes the outcome variable for property i in week t. The dependent variable is either the number of buyers visiting the property during the week or the composition of visiting buyers measured through the upper and lower bounds of their self-indicated price ranges. The upper price bound is defined as the logarithm of the average upper bound reported by buyers visiting property i in week t, and the lower bound is defined analogously.

Up_{it} and $Down_{it}$ capture treatment status. $Up_{it} = 1$ for observations after a property experiences an upward price revision (markup), and $Up_{it} = 0$ for properties without any price revisions. Similarly, $Down_{it} = 1$ for observations after a downward price revision (markdown), while $Down_{it} = 0$ for properties with unchanged list prices. The markup and markdown specifications are estimated separately, so the two treatment indicators never enter the regression simultaneously.

The control variables X_{it} vary across specifications. When the dependent variable is the number of visits, we control for the logarithm of time-on-market. When the dependent variable measures buyers' price ranges, X_{it} includes the logarithm of the number of visits. γ_i denotes property fixed effects, which absorb time-invariant housing characteristics, while ϕ_t represents calendar-week fixed effects that capture aggregate market trends over time. To further account for localized market dynamics, we also estimate specifications including complex-by-week fixed effects and neighborhood-by-week fixed effects. In specifications examining buyer composition, we additionally estimate weighted least squares (WLS) regressions using the number of visits as analytical weights, thereby assigning greater weight

[5]Figure A2 in Appendix B illustrates list price adjustments at the property-week level and the definition of the treatment and control groups.

to property-week observations with more buyer interactions. Standard errors are clustered at the apartment-complex level.

4.1.3 Continuous DID Specification

In the canonical DID model, we define dummy variables for price revisions, thereby treating all revisions equally. However, price revisions differ substantially in magnitude. Moreover, the size of a price revision is itself a seller's choice and may be endogenous: larger revisions may reflect sellers' private information, stronger selling urgency, accumulated lack of buyer interest, or beliefs about the degree of initial mispricing. Ignoring this intensive-margin variation may therefore mask heterogeneous responses in buyer attention. For this reason, we also employ a continuous DID specification following Callaway et al. (2024) to examine whether the magnitude of a list price revision affects outcome variables.

The identifying assumption is a generalized parallel-trends condition: absent the price revision, listings with different revision magnitudes would have followed comparable trends in outcome variables, conditional on the included fixed effects and controls. Under this assumption, the continuous DID specification identifies dose-specific treatment effects associated with different revision magnitudes. Interpreting differences across revision magnitudes as causal dose-response effects further requires that selection into larger or smaller revisions is not driven by unobserved factors that would have generated systematically different counterfactual trends in visits.

We define a variable $Ratio_i$ for price revision magnitudes, which is obtained by dividing the absolute value of the difference between the price before and after the revision by the pre-revision price. For properties without price revisions, $ratio_i$ is equal to 0. The continuous DID specification adds the interaction term between $ratio_i$ and treatment variables. The specification is as follows:

$$Y_{it} = \alpha + \beta Ratio_i * \{Up_{it}, Down_{it}\} + \lambda X_{it} + \gamma_i + \phi_t + \mu_{it}. \tag{10}$$

4.1.4 Event Study

The key assumption for identifying the Average Treatment Effect on the Treated (ATT) in the DID specification is that the parallel trends assumption holds. This assumption allows us to interpret the control group as a valid counterfactual for the treatment group. While the parallel trends assumption is inherently untestable, we can gain confidence in its validity if the treatment and control groups exhibit similar pre-trends before revisions. To check this, we conduct event studies to assess the parallel pre-trends and explore the dynamic effects of list price revisions. Specifically, we estimate the following two equations:

$$Y_{it} = \alpha + \sum_{\tau=-7, \tau\neq-1}^{\tau=7} \beta_\tau \mathbb{I}(T_i - t = \tau) + \lambda X_{it} + \gamma_i + \rho_n \phi_t + \mu_{it}, \tag{11}$$

$$Y_{it} = \alpha + \sum_{\tau=-7,\tau\neq-1}^{\tau=7} \beta_\tau Ratio_i * \mathbb{I}(T_i - t = \tau) + \lambda X_{it} + \gamma_i + \rho_n \phi_t + \mu_{it}. \quad (12)$$

T_i denotes the week in which house i has a list price revision. $\rho_n \phi_t$ denotes the neighborhood*week fixed effect. We retain observations for 7 periods before and after the event and use the period before the event as the baseline period, which is omitted by the specification.

4.2 Empirical Results on Visitors Composition

4.2.1 Baseline Results

We first report the staggered DID results in Table 5. The estimates compare properties that experience only upward or downward price revisions with properties in the same complex or neighborhood that do not revise their prices during the same period. Panel A shows that, following an upward list price revision, the average stated price preference of visiting buyers increases. The upper bound of buyers' stated price-preference range rises by about 0.7%–0.8%, while the lower bound increases by about 0.3%–0.5%. In contrast, Panel B shows that, following a downward revision, the average stated price preference of visiting buyers declines. The upper bound falls by about 0.6%–0.8%, and the lower bound decreases by about 0.4%–0.5%.

The corresponding event-study estimates, shown in Figure 4, support this interpretation. We find no clear evidence of differential pre-trends in buyers' stated price ranges before the revision, lending support to the parallel-trends assumption. Following an upward revision, both the upper and lower bounds increase significantly, while both decline following a downward revision. The post-revision effects are relatively persistent, with only a modest attenuation in some specifications.

Taken together, these results suggest that list prices play an important role in shaping the composition of visiting buyers. Holding the property's characteristics fixed, a price revision can reposition the listing across buyer price segments. An upward revision attracts buyers with higher stated price preferences and discourages those searching in lower price ranges, whereas a downward revision draws in buyers from lower price segments. The observed changes in visiting buyers' preferred price ranges therefore support the price-segmentation mechanism: list price revisions affect not only the attractiveness of a listing but also the composition of buyers who consider it.

4.2.2 Effects of List Price Revision Intensity

Because the magnitude of price revisions varies across listings, the staggered DID estimates may mask heterogeneity in treatment intensity. We therefore estimate a continuous DID specification using the size of the list price revision as the treatment intensity. The results are reported in Table 6. The coefficients can be interpreted as elasticities of visiting buyers' stated price preferences with respect to the magnitude of the list price revision.

For list price markups, we estimate elasticities of 0.20–0.23 and 0.15–0.16 for the upper and lower bounds of buyers' stated price ranges, respectively. For markdowns, the corresponding elasticities are 0.21–0.26 and 0.11–0.13. The continuous event-study estimates in Figure 5 show a similar pattern. There is little evidence of differential pre-trends, while larger upward revisions are followed by larger increases in buyers' stated price ranges and larger downward revisions by larger declines.

Two patterns are worth noting. First, consistent with the staggered DID results, price revisions have larger effects on the upper bound than on the lower bound of buyers' stated price preferences. Second, the estimated elasticities are substantially below one, indicating that buyer composition adjusts less than proportionally to changes in list prices. Nevertheless, the response is systematically related to revision intensity: larger price increases attract buyers with higher stated price preferences, while larger price reductions shift the visiting pool toward lower price segments. This provides further evidence that list price revisions reposition properties across segmented buyer demand.

4.3 Empirical Results on Number of Visits

Because sale probabilities are typically closely related to buyer visits, we next examine whether the higher sale hazards following list price revisions are accompanied by an increase in buyer attention. Using the same DID and event-study designs, we next examine how list price revisions affect buyer attention, measured by the number of weekly visits. The DID estimates in Table 7 show that both markdowns and markups increase buyer visits. On average, a markdown raises weekly visits by about 0.6, while a markup increases weekly visits by about 0.16 to 0.21. Thus, even upward price revisions are associated with more buyer attention, despite the higher list price. The event-study estimates in Figure 6, however, show that the dynamics differ sharply between the two types of revisions. Markdowns generate an immediate and persistent increase in visits, whereas the positive effect of markups is short-lived: visits rise in the week of the revision and the following week, but the effect quickly fades thereafter.

These results are useful to compare with Han and Strange (2016), who document a negative relationship between list prices and the number of bidders. Con-

Table 5: List Price Revisions and Buyer Composition

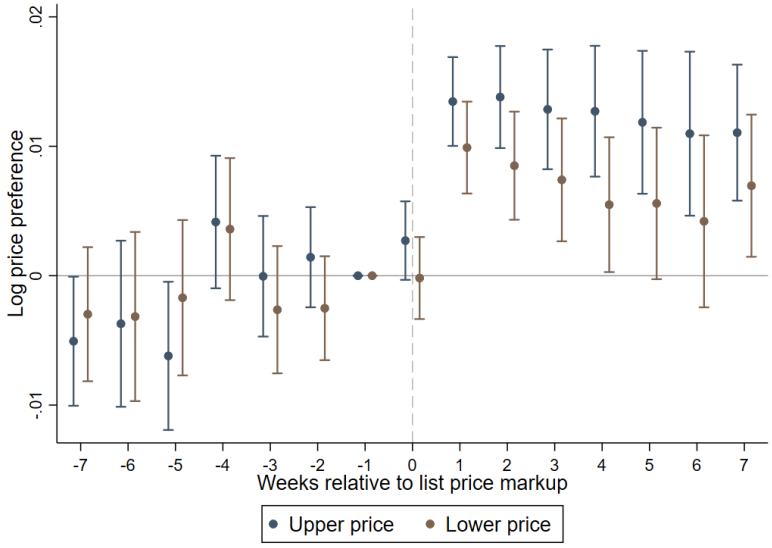
	log(upper price)			log(lower price)		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Upward list price revision						
Markup	0.008***	0.008***	0.007***	0.005***	0.005***	0.003**
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)
log(number of visits)	0.006***	0.006***	0.006***	0.006***	0.006***	0.006***
	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.001)
N	378,886	378,685	306,134	378,886	378,623	306,134
R^2	0.941	0.944	0.964	0.931	0.934	0.958
Property F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighborhood × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓
Panel B. Downward list price revision						
Markdown	-0.007***	-0.008***	-0.006***	-0.004***	-0.005***	-0.004**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
log(number of visits)	0.007***	0.007***	0.007***	0.007***	0.007***	0.007***
	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.001)
N	393,095	392,830	320,172	393,095	392,830	320,172
R^2	0.942	0.945	0.964	0.932	0.935	0.958
Property F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighborhood × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓

Notes: These results are estimated using the WLS estimator of equation (9), where the analytical weight is the number of visits. The dependent variables are the log average upper and lower bounds of the price preferences of buyers at the house-week level. Panel A reports estimates for properties with a single upward list price revision, and Panel B reports estimates for properties with a single downward list price revision. The control group consists of properties without any list price revision during the listing period. Standard errors are clustered at the apartment-complex level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

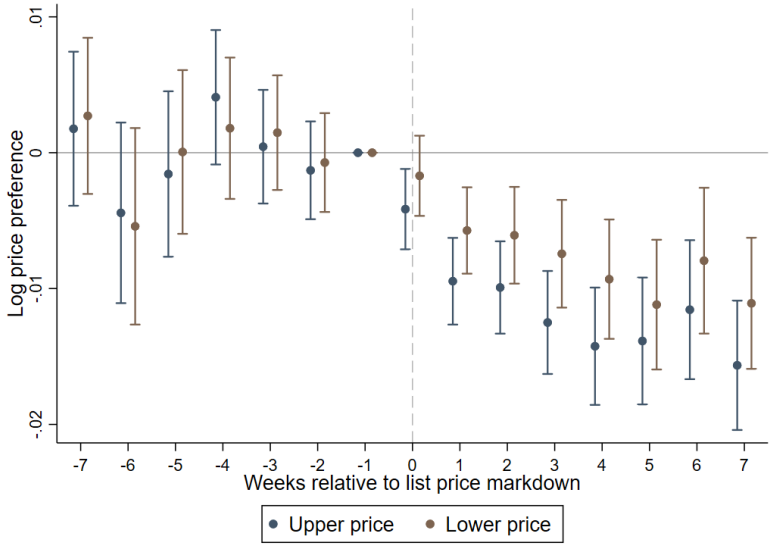
Table 6: List Price Change Intensity and Buyer Composition

	log(upper price)			log(lower price)		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Upward list price revisions						
Markup ratio	0.229***	0.228***	0.204***	0.161***	0.165***	0.149***
	(0.023)	(0.023)	(0.034)	(0.024)	(0.025)	(0.036)
log(number of visits)	0.007***	0.006***	0.006***	0.006***	0.006***	0.006***
	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.001)
N	378,886	378,623	306,134	378,886	378,623	306,134
R^2	0.941	0.944	0.964	0.931	0.934	0.958
Property F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighborhood × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓
Panel B. Downward list price revisions						
Markdown ratio	-0.250***	-0.263***	-0.210***	-0.119***	-0.134***	-0.115**
	(0.040)	(0.040)	(0.052)	(0.042)	(0.042)	(0.055)
log(number of visits)	0.007***	0.007***	0.007***	0.007***	0.007***	0.007***
	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.001)
N	393,095	392,830	320,172	393,095	392,830	320,172
R^2	0.942	0.945	0.964	0.932	0.935	0.958
Property F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighborhood × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓

Notes: These results are estimated using the WLS estimator of equation (10), where the analytical weight is the number of visits. The dependent variables are the log average upper and lower bounds of the price preferences of buyers at the house-week level. Panel A reports estimates for properties with a single upward list price revision, and Panel B reports estimates for properties with a single downward list price revision. The control group consists of properties without any list price revision during the listing period. Standard errors are clustered at the apartment-complex level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.



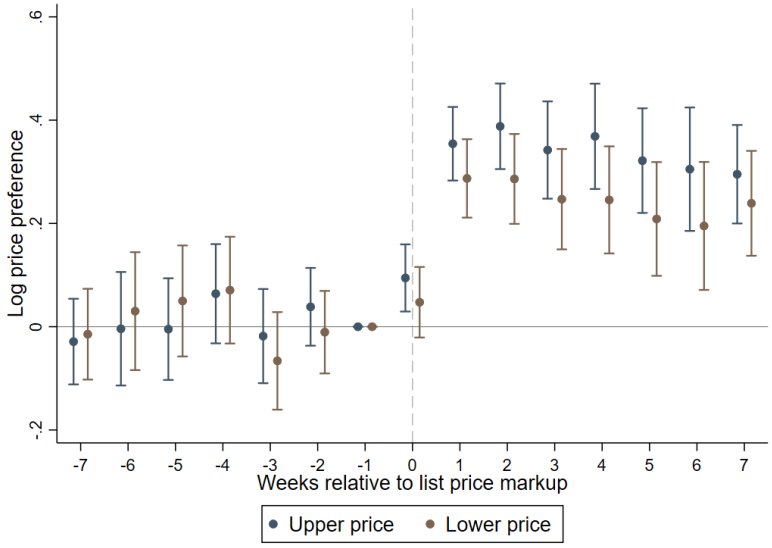
(a) Upward revision



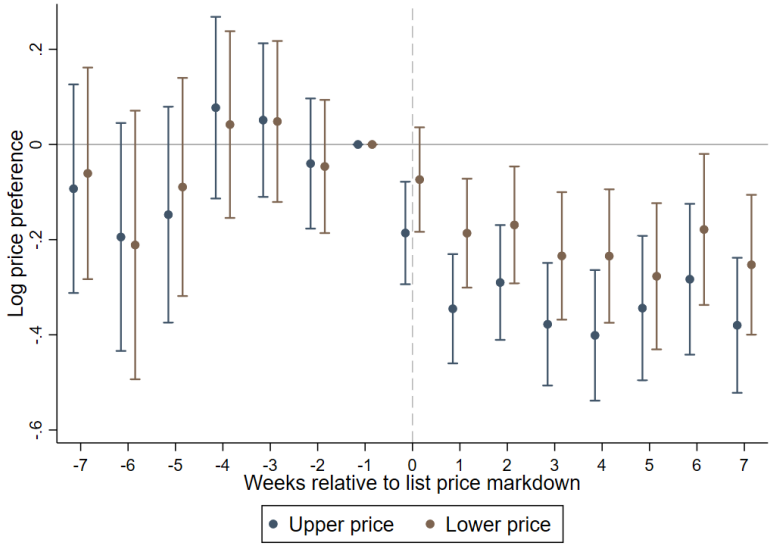
(b) Downward revision

Figure 4: The Dynamic Effects of List Price Revisions

Notes: This figure presents coefficient plots of equation (11) with a 95% confidence level. The top figure shows the dynamic effect of an increase in the list price, and the bottom figure shows the dynamic effect of a decrease in the list price. The dependent variables are the upper and lower bounds of the buyer's average price range.



(a) Upward revision



(b) Downward revision

Figure 5: The Dynamic Effects of the List Price Revision Magnitude

Notes: This figure presents coefficient plots of equation (12) with a 95% confidence level. The top figure shows the dynamic effect of an increase in the list price, and the bottom figure shows the dynamic effect of a decrease in the list price. The dependent variables are the upper and lower bounds of the buyer's average price range.

Table 7: List Price Revisions and Number of Visits

	Number of visits					
	Markup			Markdown		
	(1)	(2)	(3)	(4)	(5)	(6)
Price revision	0.214***	0.182***	0.162***	0.599***	0.604***	0.613***
	(0.018)	(0.018)	(0.023)	(0.016)	(0.016)	(0.021)
log(TOM)	-0.072***	-0.081***	-0.022	-0.004	-0.023*	0.008*
	(0.013)	(0.013)	(0.018)	(0.013)	(0.013)	(0.018)
N	523,668	523,506	451,252	535,999	535,820	463,135
R^2	0.565	0.581	0.699	0.567	0.584	0.698
Property F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighborhood × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓

Notes: The dependent variable is the number of visits at the property-week level. Columns (1)–(3) report estimates for upward list price revisions, and columns (4)–(6) report estimates for downward list price revisions. Standard errors are clustered at the apartment-complex level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

sistent with this standard demand channel, we find that lowering the list price attracts more visitors. However, we also find that raising the list price can temporarily increase buyer attention. This result is less consistent with a purely static demand interpretation, but it is consistent with the price-segmentation mechanism discussed above. When sellers raise the list price, the property may enter a higher price segment and become visible or relevant to a new group of buyers, generating a short-run increase in visits.

We provide further evidence for this mechanism in Figure 7. The figure plots the weekly number of visitors whose stated price range did not include the initial list price but does include the revised list price. Before the revision, properties that later revise their list prices do not receive substantially more visits from these buyers than comparable properties without price revisions, especially in the case of markdowns. Once the revised list price falls within these buyers' stated price ranges, however, these properties immediately attract more visits from this group. This pattern supports the interpretation that list price revisions affect buyer attention partly by moving listings into different price segments and changing the set of buyers for whom the property becomes relevant.

4.4 Robustness Checks

We conduct several robustness checks to strengthen our baseline estimates.

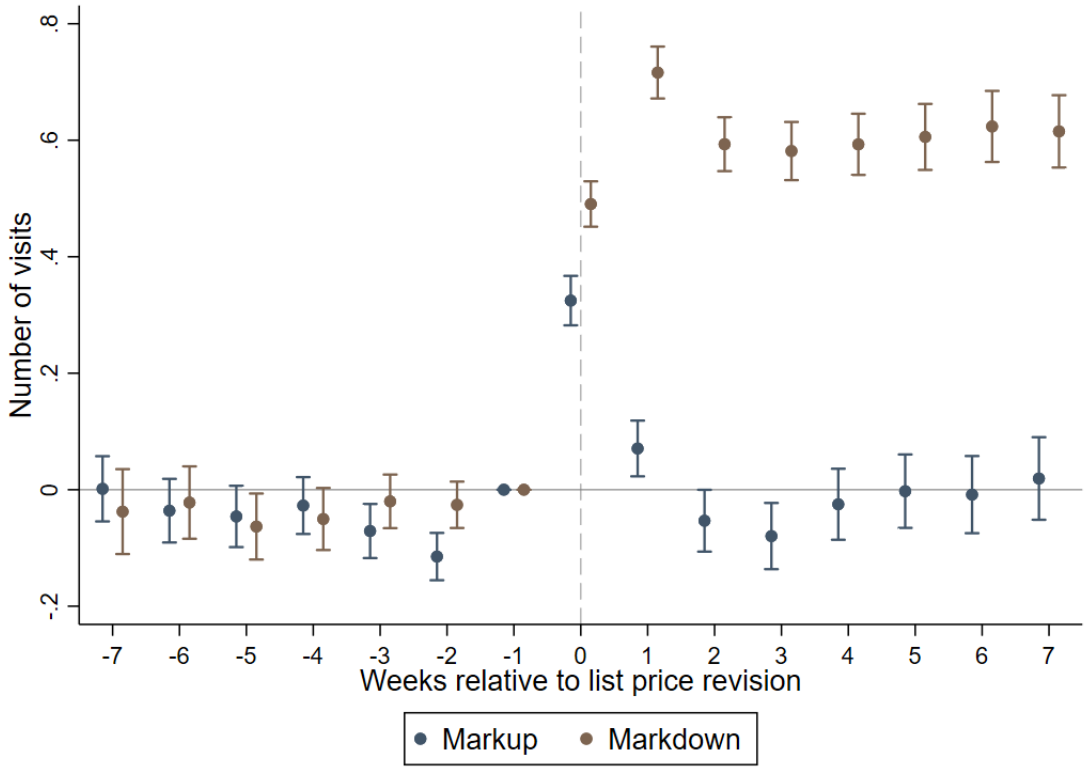


Figure 6: Dynamic Effects of List Price Revisions on Buyer Visits

Notes: This figure presents coefficient plots of equation (11) with a 95% confidence level. It shows the number of visits at house-week level change after upward and downward list price revision. The horizontal axis is the weeks relative to the list price revisions.

4.4.1 Diagnostics for Heterogeneous Treatment Effects

Recent advances in the DID literature have emphasized that, under staggered treatment adoption, the conventional two-way fixed effects (TWFE) estimator may produce biased estimates when treatment effects are heterogeneous across cohorts or over time (Callaway and Sant'Anna, 2021; de Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021). In particular, the TWFE estimator can be expressed as a weighted average of cohort-specific treatment effects, where some weights may be negative. Such negative weights can bias the overall estimate and potentially lead to misleading conclusions.

To address the issue of potentially negative weights in two-way fixed effects estimators, this paper adopts a stacked difference-in-differences (DID) approach following Sun and Abraham (2021). This approach allows for a transparent aggregation of cohort-specific treatment effects and avoids the undesirable weighting schemes inherent in standard TWFE specifications. The choice of this method is also guided by practical considerations. Alternative estimators, such as the group-time average treatment effects framework of Callaway and Sant'Anna (2021), require a balanced panel structure, which is not satisfied in the present setting. In addition, imputation-based estimators such as Borusyak et al. (2024) are compu-

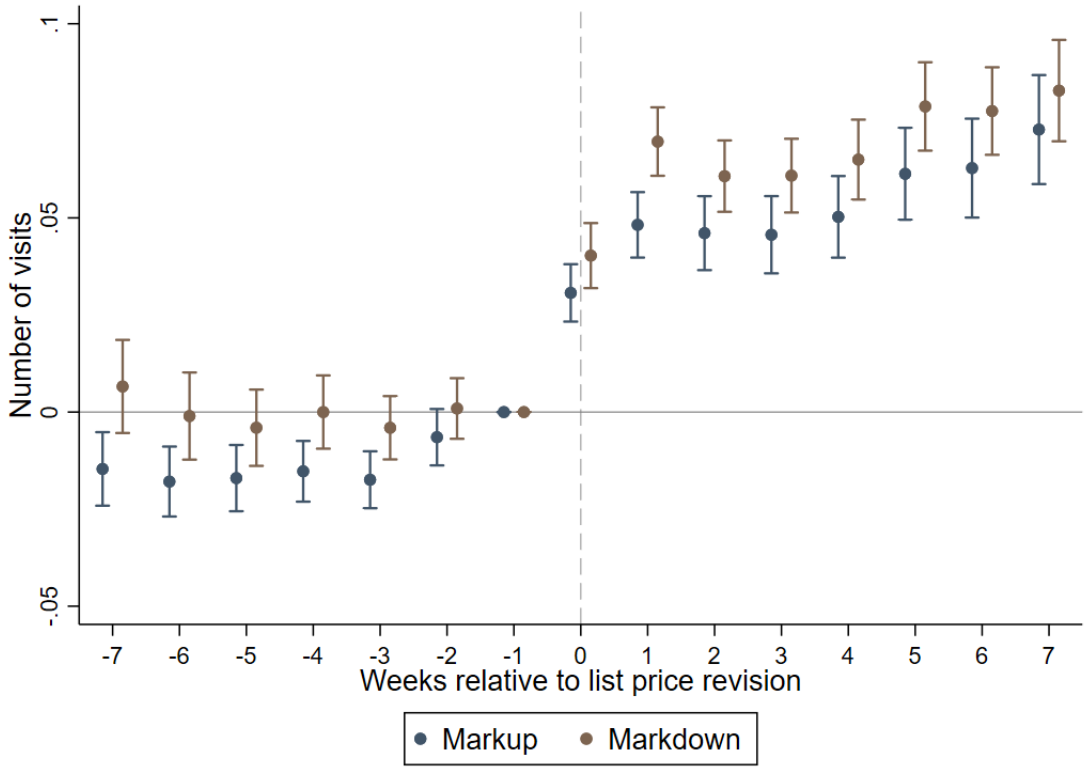


Figure 7: Dynamic Effects of List Price Revisions on Visits from New Price Segments

Notes: The figure shows the coefficients of an event study on the impact of both upward and downward list price revisions on the weekly number of visitors per property that, upon registration, indicated a price range that did not contain the initial list price, but does contain the revised list price.

tationally infeasible in our context due to the large sample size and convergence issues. The stacked DID approach therefore provides a tractable and reliable implementation in this setting.

Figure A6 and Figure A7 present the dynamic treatment effects of list price revisions on buyers' stated price ranges and visits, respectively, using the interaction-weighted estimator of Sun and Abraham (2021). The figures show no evidence of systematic pre-trends, supporting the identifying assumption. Following the revision, both outcomes respond immediately, and the dynamic patterns are highly consistent with those obtained from the baseline specifications, suggesting that accounting for heterogeneous treatment timing does not materially alter the conclusions.

4.4.2 The Multiple Price Revisions

Since the staggered DID strategy retains only samples with either one price adjustment or no price adjustments, the estimation results may be selective and limited. We next consider sellers' multiple price revisions using the entire sample includ-

ing multiple price revisions as a robustness check. We calculate the cumulative numbers of upward and downward list price revisions for each property by week in order to capture the effects of repeated price adjustments. The dependent variables remain the average price preferences of visiting buyers and the number of weekly visits, respectively. We estimate the following TWFE model:

$$Y_{it} = \alpha + \beta_1 CumMarkup_{it} + \beta_2 CumMarkdown_{it} + \lambda X_{it} + \gamma_i + \phi_t + \epsilon_{it}, \quad (13)$$

where Y_{it} denotes the outcome variable for property i in week t, including the average upper bound, lower bound, or the number of visits. $CumMarkup_{it}$ and $CumMarkdown_{it}$ measure the cumulative numbers of upward and downward list price revisions up to week t, respectively. X_{it} indicates the logarithm of visits when average upper bound, lower bound price serve as dependent variables. γ_i denotes property fixed effects, which absorb time-invariant property characteristics, including unobserved housing attributes, while ϕ_t denotes week fixed effects that capture aggregate housing market conditions over time. ϵ_{it} is the error term.

The results presented in Table A3 report the relationship between cumulative list price revisions and buyer composition and visits. We find that cumulative markups are positively associated with both the upper and lower bounds of visitors' self-indicated price ranges, suggesting that repeated upward price revisions attract buyers with systematically higher price preferences. In contrast, cumulative markdowns are associated with lower buyer price ranges, indicating that downward price revisions reposition listings toward lower-price buyer segments. At the same time, both cumulative markups and markdowns significantly increase buyer visits, although the effect of markdowns is substantially larger in magnitude. These results are highly consistent with our baseline DID estimates in Table 5 and 7 based on the sample of properties with either a single price revision or no price revision. This suggests that, although the DID specification relies on a selected subsample, the sample restriction does not materially affect our main conclusions.

5 Conclusion

In housing markets, sellers present their properties on a platform and announce list prices to attract prospective buyers. While housing characteristics are largely fixed, sellers can adjust list prices to influence the sales process. This paper shows that list price revisions attract different groups of buyers to a listed property. Specifically, upward revisions are followed by more visits from buyers interested in higher-priced houses, whereas markdowns attract visitors who previously ignored the property because it fell outside their preferred price range. These findings suggest that list prices help position listings in different segments of the housing market.

Our data come from the Chinese housing market and reveal a high frequency of list price revisions. More than 60% of sellers revise their list price at least once, a share substantially higher than that documented for the UK and the Netherlands (de Wit and van der Klaauw, 2013; Merlo et al., 2015). The substantial proportion of upward revisions also provides a valuable setting for studying the effects of both upward and downward price changes.

We find that both upward and downward list price revisions increase the number of visitors immediately after the change, but with important differences. The increase in visits is smaller and more short-lived for upward revisions. A similar pattern emerges for sale probabilities. Downward revisions are associated with a substantially higher probability of sale, while upward revisions are also associated with higher sale probabilities relative to properties without price revisions. However, the effect of upward revisions is smaller and less persistent. Sales prices decline after downward revisions but increase after upward revisions, which may explain why upward revisions can be an attractive sales strategy.

To interpret these findings, we develop a model of matching in the housing market in which heterogeneous buyers follow a Weitzman (1979)-type sequential search strategy and visit the houses that appear most attractive based on the information provided by the platform. buyers differ in their preferred price and quality ranges and therefore concentrate on subsets of listings. A change in the list price can bring a property to the attention of buyers who had previously ignored it, explaining why list price revisions affect visits and why the effects differ between upward and downward revisions.

The model also predicts that the visiting rate of listed houses is initially high but declines over time, as demand from buyers already active when the house was listed gradually dries up. This negative duration dependence gives sellers an incentive to revise the list price after some time in order to attract buyers who previously ignored the property because it was outside their price range. Our model is related to stock-flow matching, but it predicts a gradual decline in visiting and sale rates over time rather than a sharp dichotomy between an immediate stock effect and a lower permanent flow effect.

These conclusions are important for understanding the microstructure of housing markets, especially the interaction between sellers and buyers and the role of list prices. We document in greater detail than previous studies how list prices attract selected groups of prospective buyers and steer housing search. Modeling housing search in line with Weitzman (1979) helps explain why buyers concentrate on specific market segments and why buyer arrival rates decline over time. The latter clarifies why many sellers revise list prices after some time, while the former shows how such revisions can bring a property to the attention of different groups of buyers. These insights help open the black box of the housing-market matching process, which remains one of the central challenges in real estate economics.

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Appendix

A Proof

A.1 Proof of Lemma 1

Proof. The first-ranked house is visited with probability one. For $m \geq 2$, buyer i visits the house ranked m if and only if she has not stopped after visiting ranks $1, \ldots, m-1$. With recall, this requires the highest realized utility among previously visited houses to be below the reservation index of the house ranked m:

$$\max_{s<m} u_{ik_s} < r_{ik_m}.$$

Equivalently,

$$u_{ik_s} < r_{ik_m} \qquad \text{for all } s < m.$$

Since post-visit shocks are independent across houses, the visiting probability is

$$\Pr^{\text{visit}}_{ik_m}(m) = \prod_{s=1}^{m-1} \Pr(u_{ik_s} < r_{ik_m}).$$

For monotonicity, define

$$P_m = \Pr^{\text{visit}}_{ik_m}(m).$$

To visit rank $m+1$, buyer i must first visit rank m and then continue searching after rank m. Hence

$$P_{m+1} = P_m \cdot \Pr(\text{continue from rank } m \text{ to } m+1 \mid V_m),$$

where V_m is the event that buyer i visits rank m.

It remains only to show that this continuation probability is strictly smaller than one. Equation (2.2) defines the reservation threshold z^*. Since $c > 0$, we have $z^* < \bar{z}$. Given

$$r_{ik_m} = v_{ik_m} + z^* - p_{k_m} \qquad \text{and} \qquad u_{ik_m} = v_{ik_m} + z_{ik_m} - p_{k_m},$$

the condition $z^* < \bar{z}$ implies

$$\Pr(z_{ik_m} \geq z^*) > 0,$$

and therefore

$$\Pr(u_{ik_m} \geq r_{ik_m}) > 0.$$

Since the Weitzman indices are ordered as

$$r_{ik_m} \geq r_{ik_{m+1}},$$

the event $u_{ik_m} \geq r_{ik_m}$ implies that buyer i stops after visiting rank m, before rank $m+1$ is reached. Thus,

$$\Pr(\text{continue from rank } m \text{ to } m+1 \mid V_m) < 1.$$

Therefore,

$$P_{m+1} < P_m.$$

This proves that the visiting probability is strictly decreasing in rank. □

B Additional Tables and Figures

Table A1: Summary Statistics: Housing Characteristics

	Full sample		Selected sample	
	N	Mean	*N*	Mean
Initial list price (10,000 RMB)	393,029	391.31	191,219	382.59
Final list price (10,000 RMB)	393,029	393.66	191,219	383.72
Size (m^2)	386,868	91.84	189,253	90.05
Floor	386,280	7.32	188,228	7.37
Building height	388,288	13.55	189,391	13.53
School	393,029	0.75	191,219	0.74
Subway	393,029	0.71	191,219	0.70
Number of bedrooms	392,020	2.13	190,891	2.09
Number of living rooms	389,978	1.21	190,111	1.20
Number of bathrooms	387,949	1.26	189,249	1.24
Construction year	357,396	2000.32	174,054	2000.45
Number of visits	393,029	8.07	191,219	8.63

Notes: This table compares housing characteristics in the full sample and the selected sample. The full sample includes all properties in the data, whereas the selected sample is restricted to properties with zero or one list price revision during the listing period. Floor refers to the floor level of the apartment, while building height indicates the total number of floors in the building. School and subway indicate whether there is a school or subway station near the property.

Table A2: Summary Statistics of Buyer Preference and Visiting

Variable	Mean	Std. Dev.	Min.	Max.	Observations
Upper price	409.68	224.97	100	1600	570,581
Lower price	329.33	174.30	50	1100	570,581
Upper floor area	99.47	37.85	42	280	570,581
Lower floor area	77.28	30.10	30	200	570,581
Upper bedrooms	2.38	0.77	1	6	570,581
Lower bedrooms	1.93	0.68	1	6	570,581
Number of total visits	5.47	4.984	1	30	570,581

Notes: This table reports summary statistics for buyers' stated search criteria and visiting behavior. The sample consists of buyers who made at least one property visit. Upper price and lower price refer to the bounds of buyers' preferred price ranges, measured in 10,000 CNY. Upper floor area and lower floor area refer to the bounds of preferred floor area, measured in square meters. Upper bedrooms and lower bedrooms refer to the bounds of preferred number of bedrooms. Number of total visits denotes the number of property visits made by a buyer.

Table A3: The Impact of Cumulative Price Revisions

	log(upper price)		log(lower price)		Number of visits	
	(1)	(2)	(3)	(4)	(5)	(6)
Cum. markups	0.013***	0.009***	0.011***	0.007***	0.046***	0.043***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.003)	(0.003)
Cum. markdowns	-0.007***	-0.006***	-0.006***	-0.005***	0.180***	0.169***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.004)
log(number of visits)	0.007***	0.006***	0.007***	0.007***		
	(0.000)	(0.000)	(0.000)	(0.000)		
N	1,087,855	1,024,961	1,087,855	1,024,961	6,692,471	6,657,554
R^2	0.935	0.953	0.924	0.945	0.354	0.394
Property F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Complex × Week F.E.		✓		✓		✓

Notes: Columns (1)–(4) examine how cumulative price revisions are associated with the price preferences of visiting buyers, measured by the average upper and lower bounds of visitors' self-indicated price ranges. Columns (5)–(6) examine the relationship between cumulative price revisions and the number of visits. The estimation uses property-week level observations. Complex-by-week fixed effects absorb local market conditions at a highly granular level. Standard errors are clustered at the apartment-complex level. *** p<0.01, ** p<0.05, * p<0.1.

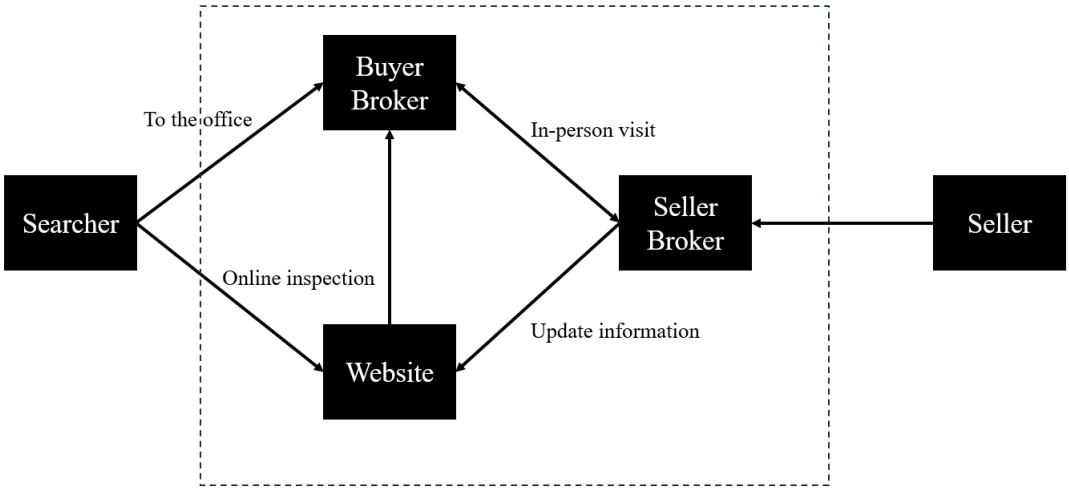


Figure A1: Schematic Illustration of Platform Operations

Notes: This diagram describes how the platform operates and how buyers and sellers are matched. Buyers must visit properties through a real estate agent. Beforehand, buyers can browse properties on the platform to identify suitable ones. Information exchange between buyers and sellers is achieved through both the website and brokers. In the figure, the dotted line represents the real estate platform.

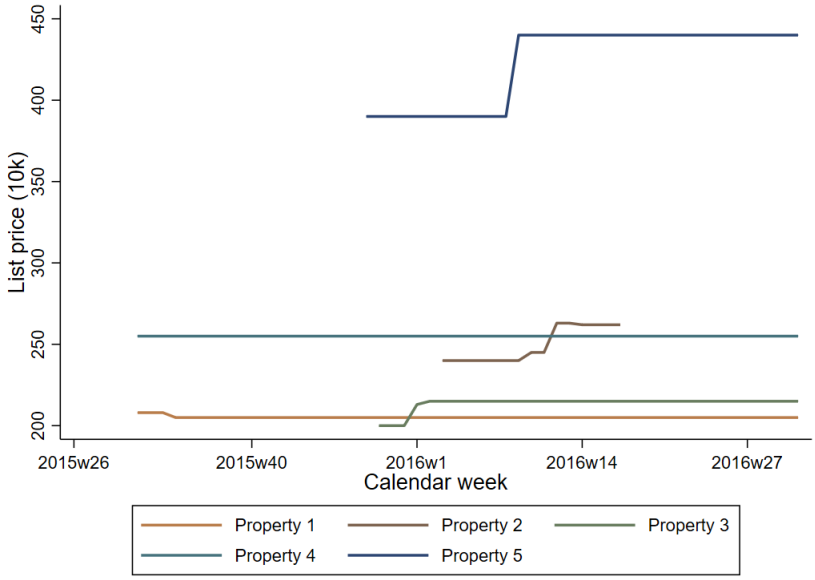


Figure A2: Illustration of List Price Dynamics and Treatment Classification

Notes: This figure illustrates the dynamics of list prices and the classification of treatment and control properties. The horizontal axis represents calendar week, while the vertical axis reports the list price. Each line corresponds to a single property. Property 1 experiences a single downward price revision, and Property 5 experiences a single upward price revision; these two properties are therefore classified as treated properties. Property 3 does not experience any price revision and is classified as a control property.

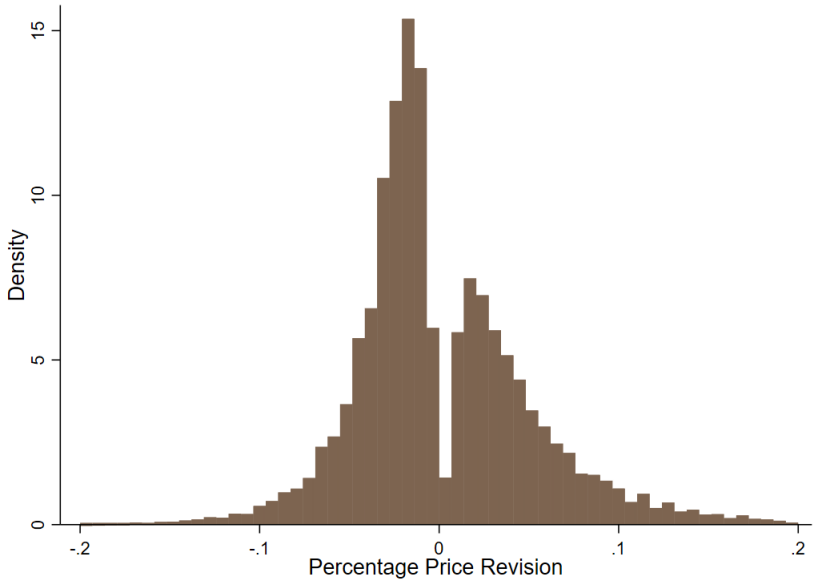


Figure A3: Distribution of the Percentage Price Revision

Notes: The figure plots the distribution of the percentage price revision, defined as $\frac{Price_{post}-Price_{pre}}{Price_{pre}}$, where $Price_{pre}$ and $Price_{post}$ denote the list prices before and after the adjustment, respectively.

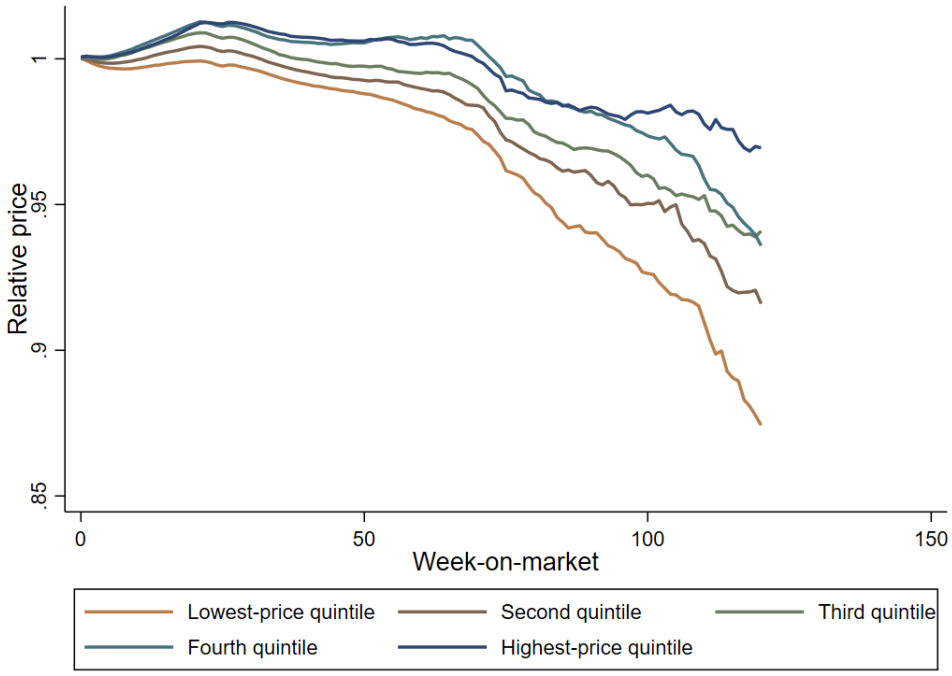


Figure A4: List Price Relative to Initial Price over Time-on-Market

Notes: The horizontal axis measures the number of weeks since the property was initially listed. The vertical axis reports the relative price, which is defined by the ratio of the property's list price in that week to its initial list price. Properties are divided into five price quintiles. Quintile 1 contains the lowest priced properties, while Quintile 5 contains the highest-priced properties.

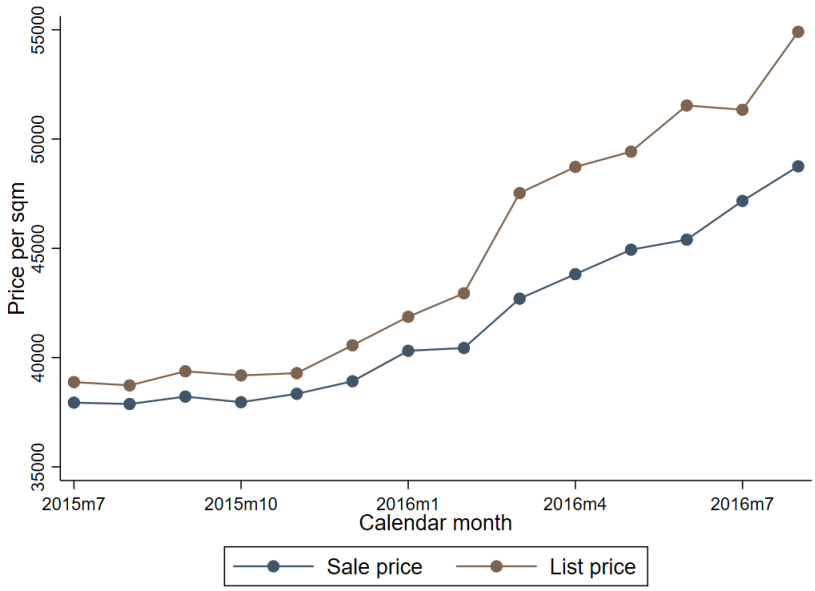
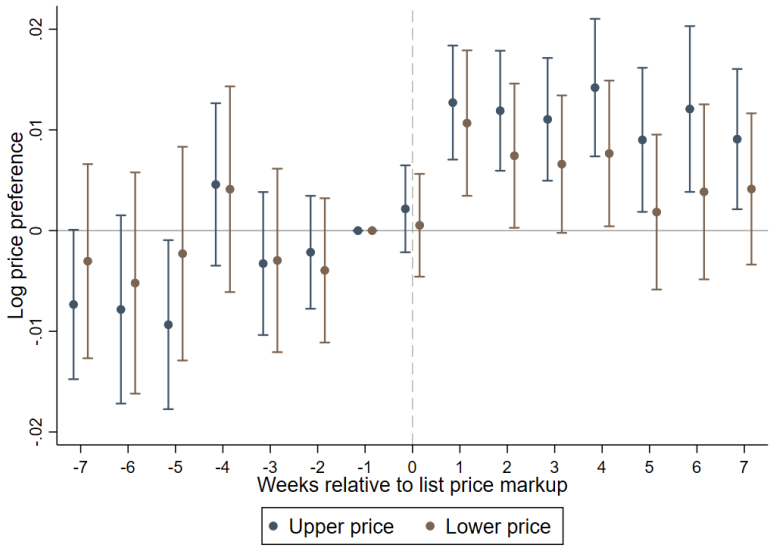
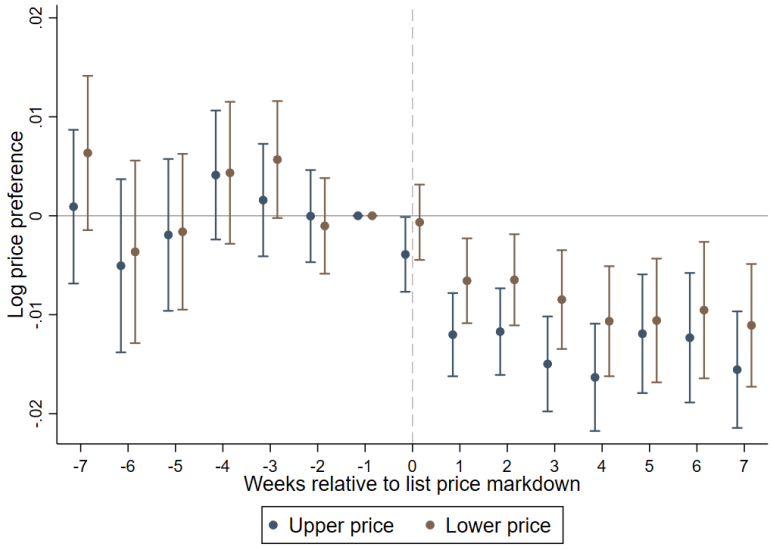


Figure A5: Price Dynamics in the Beijing Housing Market

Notes: This graph shows the change of listing and sale price over sample period. Average list and sale prices are calculated based on data used in this paper. The currency unit is CNY.



(a) Upward revision



(b) Downward revision

Figure A6: Alternative Estimates of the Effects of List Price Revisions on Buyers' Price Ranges

Notes: The figure plots event-study estimates of the effects of list price revisions on buyers' stated price ranges. The coefficients are estimated using the interaction-weighted estimator of Sun and Abraham (2021), which is robust to heterogeneous treatment timing. Panel (a) shows the effects of upward price revisions, while Panel (b) shows the effects of downward price revisions. The dependent variables are the upper and lower bounds of buyers' stated price ranges. Standard errors are clustered at apartment-complex level.

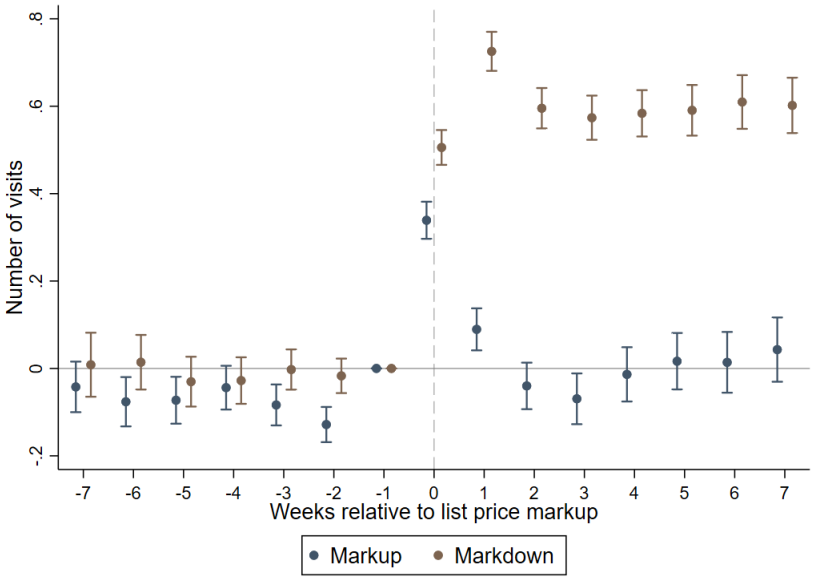


Figure A7: Alternative Estimates of the Effects of List Price Revisions on Visits

Notes: The figure plots event-study estimates of the effects of upward and downward list price revisions on the weekly number of visits. The coefficients are estimated using the interaction-weighted estimator of Sun and Abraham (2021), which is robust to heterogeneous treatment timing. Standard errors are clustered at apartment-complex level.